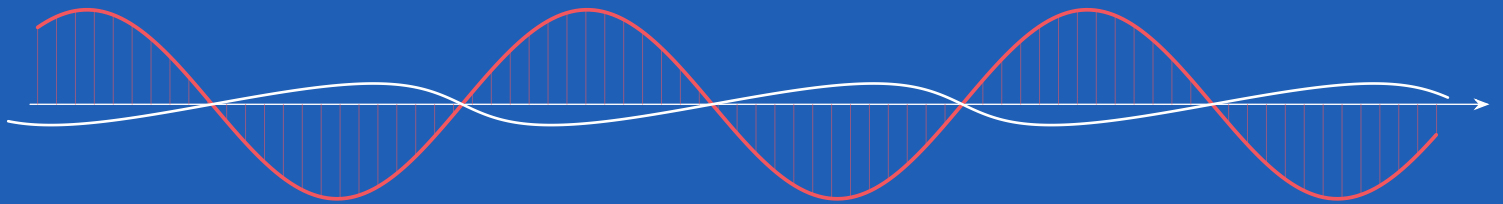


Electromagnetics

الکترومغناطیس



Ferdowsi University of Mashhad

دانشگاه فردوسی مشهد

Instructor: سید محمد سعید ماجدی

Compiled and edited by: Aidin Shekari

Electromagnetics

الکترومغناطیس

Lecture Notes

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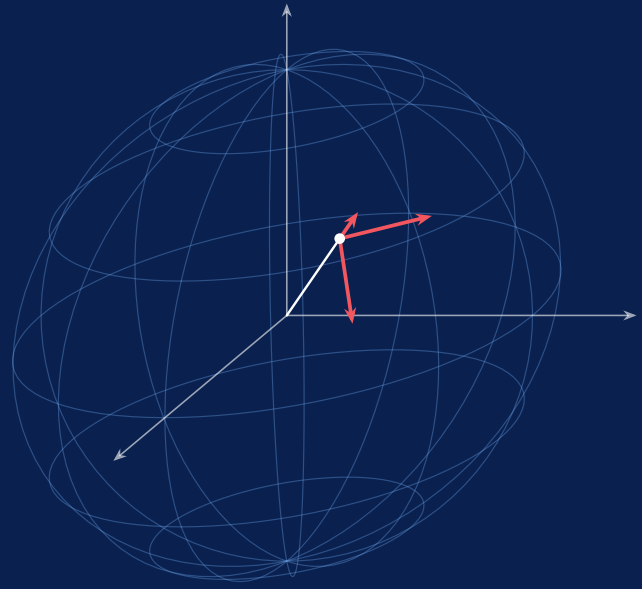
دانشگاه فردوسی مشهد

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CHAPTER

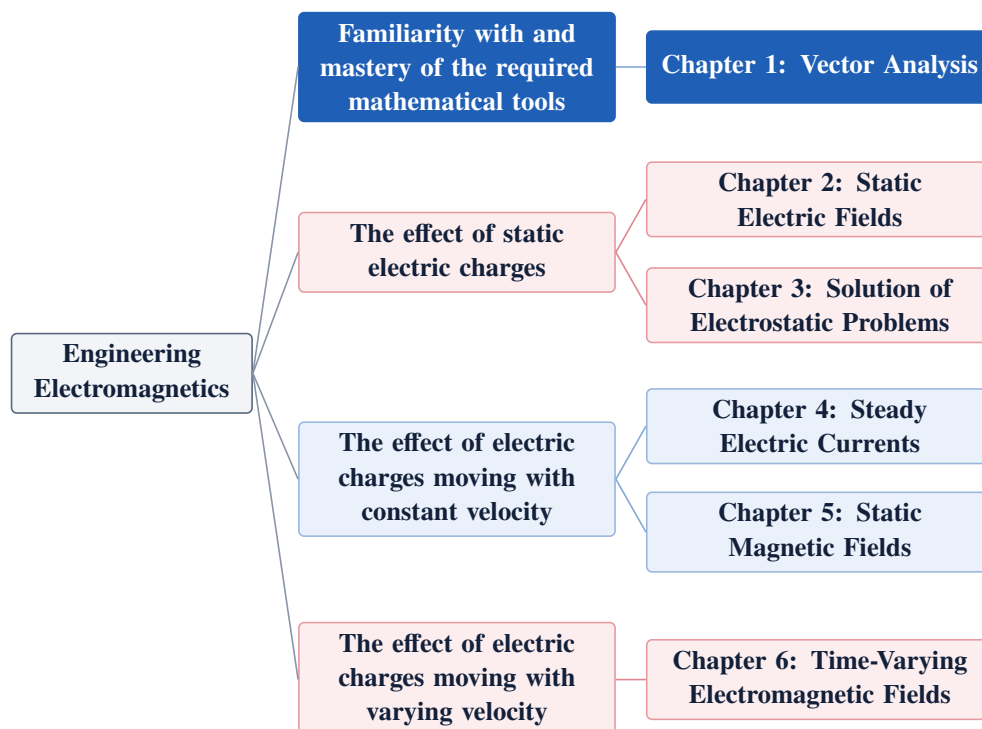
Vector Analysis

آنالیز برداری

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1.1 Course Outline

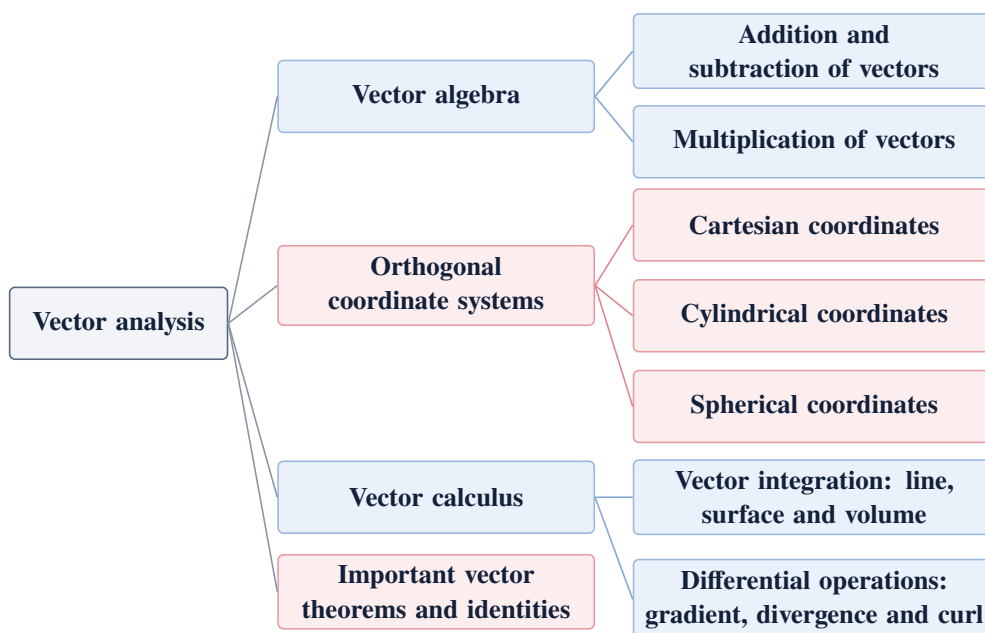


1.2 Vector Analysis

- **Vector analysis** is a mathematical tool with which the expression and understanding of electromagnetic concepts become simple.
- Quantities in electromagnetics are divided into the following two groups.
 - **Scalar:** a quantity that is completely specified by its magnitude (positive or negative). For example: charge, electric current, energy, ...
 - **Vector:** a quantity that is described by a magnitude and a direction. For example the electric and magnetic field intensity, ...
- Both of the above quantities can be functions of position and time.

Example 1-1

- In your opinion, of what type are the following quantities?
 - Temperature
 - Power
 - Force
 - Potential difference
 - Velocity
- Apart from the above, which other scalar and vector quantities in physics can you name?



- How can two vector quantities be added or subtracted graphically?
- How can vectors be multiplied? What is the meaning of this product? Where is it used?

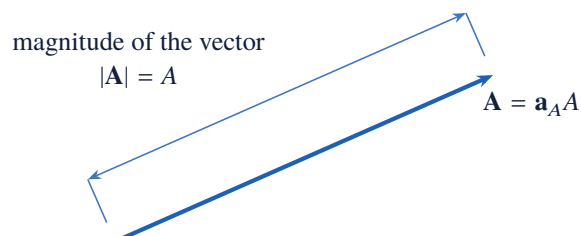
$$\mathbf{A} = 2\mathbf{a}_x + 3\mathbf{a}_y + \mathbf{a}_z \quad \mathbf{B} = \mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z \quad \mathbf{C} = -\mathbf{a}_x + 4\mathbf{a}_y - \mathbf{a}_z$$

- Find the magnitude of the projection of vector $\mathbf{A}$ on vector $\mathbf{B}$.
- What is the angle between vectors $\mathbf{A}$ and $\mathbf{B}$?
- What is the area of the triangle formed by the two vectors $\mathbf{A}$ and $\mathbf{B}$?
- Find the unit vector normal to the plane containing the two vectors $\mathbf{A}$ and $\mathbf{B}$.
- What is the volume of the parallelepiped formed by the 3 vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$?

1.3 Vector Algebra

- Every vector has a magnitude and a direction.

$$\mathbf{A} = \mathbf{a}_A A, \quad \mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A} \quad (\text{unit vector})$$



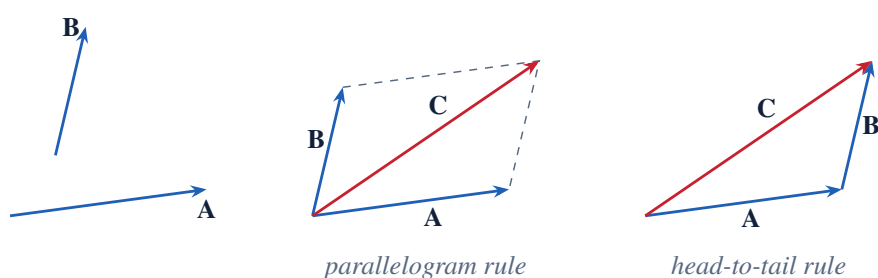
① Remark

For every vector quantity both the magnitude and the direction must be known. Leaving the direction of a vector quantity unspecified means ignoring part of the information of that quantity.

1.4 Addition of Vectors

► Adding two vectors graphically

- The **parallelogram rule**
- The **head-to-tail rule**



► Laws governing the addition of vectors

- **Commutative law**

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

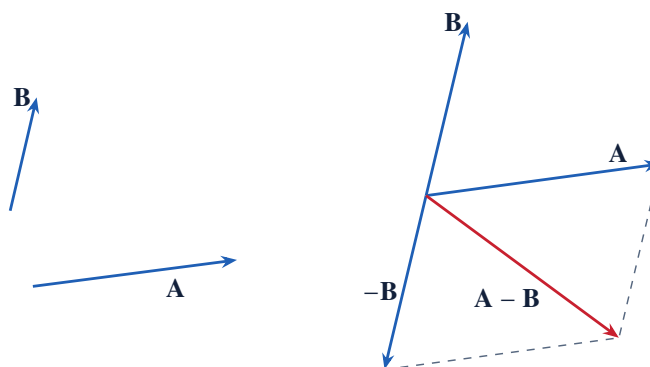
- **Associative law**

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

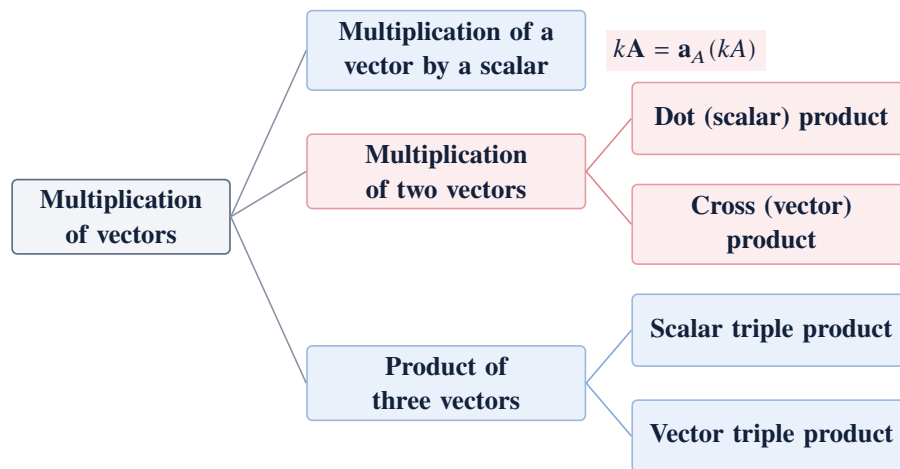
1.5 Subtraction of Vectors

► Vector subtraction

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$



1.6 Multiplication of Vectors



- Multiplication of a vector by a scalar:

$$k\mathbf{A} = \mathbf{a}_A(kA)$$

1.7 Dot or Scalar Product

≡ Definition | Mathematical definition

$$\mathbf{A} \cdot \mathbf{B} \triangleq AB \cos \theta_{AB}$$

- It equals the product of the magnitude of one vector and the projection of the other vector on the first.

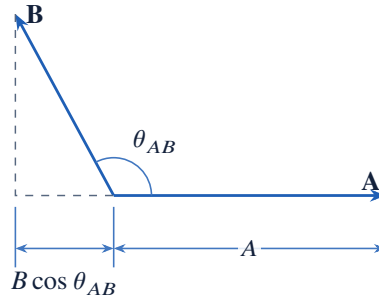


Figure 1.1 | θ_{AB} : the smaller angle between the two vectors when their tails are joined

► Properties of the dot product

- The dot product of two vectors is a scalar quantity.
- It is less than or equal to the product of their magnitudes.
- Depending on the angle between them it can be positive or negative.
- If the vectors are perpendicular to each other, it is zero.
- It is commutative and distributive.

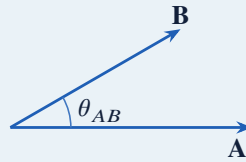
$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}, \quad \mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$$

- The dot product of a vector with itself

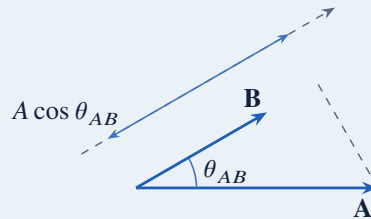
$$\mathbf{A} \cdot \mathbf{A} = A^2, \quad A = +\sqrt{\mathbf{A} \cdot \mathbf{A}}$$

Example 1-2

Using the concept of the dot product, find the projection of vector $\mathbf{A}$ on $\mathbf{B}$.



► **Solution.**



Magnitude of the projection of vector $\mathbf{A}$ on $\mathbf{B}$:

$$A \cos \theta_{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{B}$$

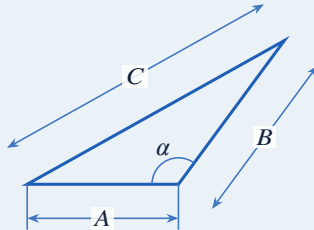
Projection of vector $\mathbf{A}$ on $\mathbf{B}$:

$$\frac{\mathbf{A} \cdot \mathbf{B}}{B} \mathbf{a}_B = \left(\frac{\mathbf{A} \cdot \mathbf{B}}{B} \right) \frac{\mathbf{B}}{B} = \frac{\mathbf{A} \cdot \mathbf{B}}{B^2} \mathbf{B}$$

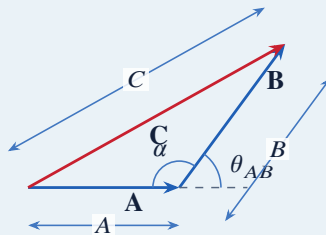
Example 1-3

Prove the law of cosines for a triangle using the concept of the dot product.

$$C = \sqrt{A^2 + B^2 - 2AB \cos \alpha}$$



► **Solution.**



$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$\begin{aligned} C^2 &= \mathbf{C} \cdot \mathbf{C} = (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} + \mathbf{B}) \\ &= \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} + 2\mathbf{A} \cdot \mathbf{B} \\ &= A^2 + B^2 + 2AB \cos \theta_{AB} \end{aligned}$$

$$\cos \theta_{AB} = \cos(180^\circ - \alpha) = -\cos \alpha$$

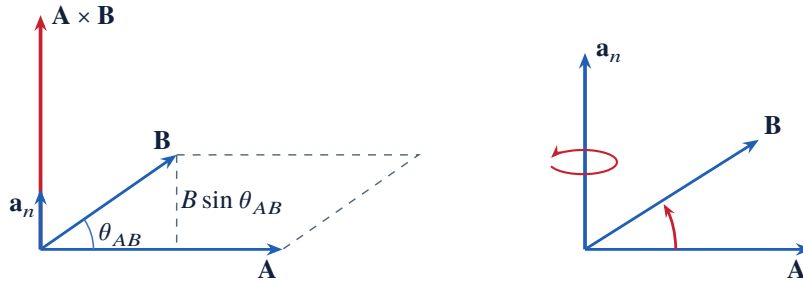
$$C^2 = A^2 + B^2 - 2AB \cos \alpha$$

1.8 Cross or Vector Product

Definition

$$\mathbf{A} \times \mathbf{B} \triangleq \mathbf{a}_n |AB \sin \theta_{AB}|$$

- The magnitude of this product equals the area of the parallelogram formed by the vectors **A** and **B**, and its direction is perpendicular to both vectors.



- Properties of the cross product

- The cross product of two vectors is a vector quantity.
- It is not commutative

$$\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}$$

- It is distributive

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$

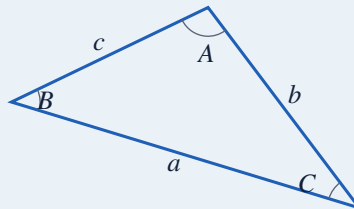
- It is not associative

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$$

Example 1-4

Prove the law of sines for a triangle using the concept of the cross product.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



► Solution.

Area of the triangle:

$$\left| \frac{1}{2} \mathbf{a} \times \mathbf{b} \right| = \left| \frac{1}{2} \mathbf{b} \times \mathbf{c} \right| = \left| \frac{1}{2} \mathbf{c} \times \mathbf{a} \right|$$

$$ab \sin C = bc \sin A = ca \sin B$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

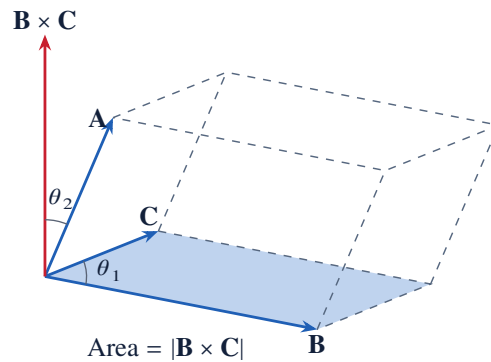
1.9 Product of Three Vectors

➤ Scalar triple product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

➤ Its magnitude equals the volume of the parallelepiped formed by the three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$.

- Area of the base: $|\mathbf{B} \times \mathbf{C}| = |BC \sin \theta_1|$
- Height: $|A \cos \theta_2|$
- Volume of the parallelepiped: $|ABC \sin \theta_1 \cos \theta_2|$



➤ Vector triple product

- Known as the back-cab rule

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

🔍 Example 1-5

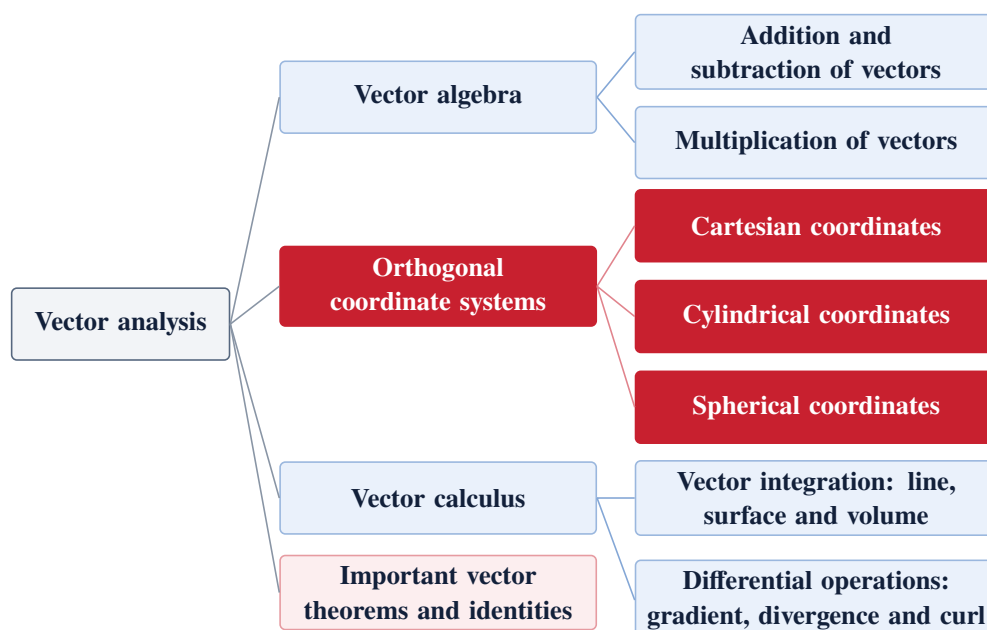
The expression $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ is equal to which of the following expressions?

- $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
- $-\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) + \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
- $-\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$
- $\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$

➤ Solution.

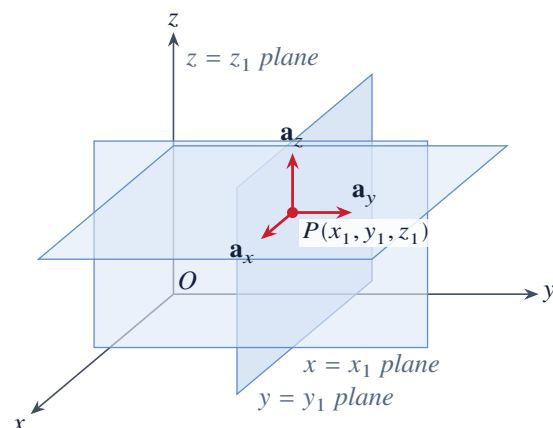
$$\begin{aligned} (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} &= -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) \\ &= -\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{B}(\mathbf{C} \cdot \mathbf{A}) \end{aligned}$$

1.10 Orthogonal Coordinate Systems



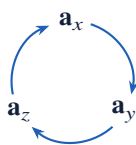
- What are the principal surfaces in the different coordinate systems?
- How are the unit vectors defined in the different coordinate systems?
- How are the position vector of a point and the distance vector between two points computed in the different coordinate systems?
- Determine the differential length, surface and volume in the different coordinate systems, for use in integration.
- Find the representation of the vector $\mathbf{A} = \mathbf{a}_r(3 \cos \phi) - \mathbf{a}_\phi 2r + \mathbf{a}_z 5$ in Cartesian coordinates.
- The position of point P in spherical coordinates is $(8, 120^\circ, 330^\circ)$. Express the position of this point in cylindrical and Cartesian coordinates.
- To describe the spatial variations of physical quantities, we must be able to describe all points of space uniquely and appropriately.
 - This requires the use of an appropriate coordinate system.
- The laws of electromagnetics do not depend on the coordinate system.
- For simplicity of the calculations, a coordinate system that suits the geometry of the problem is used.
- In three-dimensional space a point can be located at the intersection of three surfaces.
 - If these three surfaces are mutually perpendicular, we have an orthogonal coordinate system.
- The three orthogonal systems used in this course
 - Cartesian (rectangular) coordinates
 - Cylindrical coordinates
 - Spherical coordinates

1.11 Cartesian Coordinate System



- Every point $P(x_1, y_1, z_1)$ is the intersection of the three planes $x = x_1$, $y = y_1$ and $z = z_1$.
- Unit vector $\mathbf{a}_x \rightarrow$ the unit vector normal to the x -constant plane, in the positive direction
- Unit vector $\mathbf{a}_y \rightarrow$ the unit vector normal to the y -constant plane, in the positive direction
- Unit vector $\mathbf{a}_z \rightarrow$ the unit vector normal to the z -constant plane, in the positive direction
- Cross and dot products of the unit vectors

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z, \quad \mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x, \quad \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$



$$\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = 0$$

$$\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

- Representation of an arbitrary vector $\mathbf{A}$ in Cartesian coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

► **Position vector**

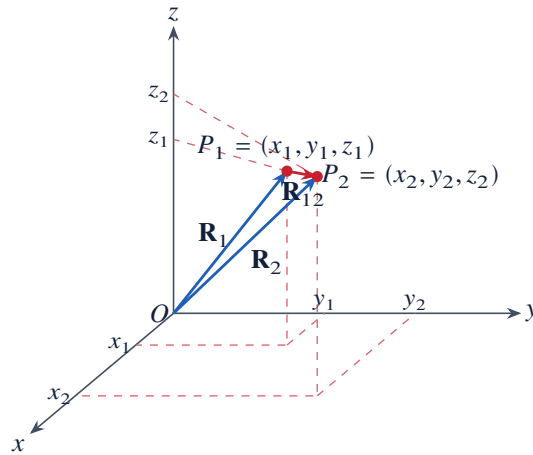
- The position vector of point P is the directed distance from the origin to P .

$$\mathbf{R}_1 = x_1 \mathbf{a}_x + y_1 \mathbf{a}_y + z_1 \mathbf{a}_z, \quad \mathbf{R}_2 = x_2 \mathbf{a}_x + y_2 \mathbf{a}_y + z_2 \mathbf{a}_z$$

► **Distance vector**

- The displacement vector from one point to another.

$$\mathbf{R}_{12} = \mathbf{R}_2 - \mathbf{R}_1 = (x_2 - x_1) \mathbf{a}_x + (y_2 - y_1) \mathbf{a}_y + (z_2 - z_1) \mathbf{a}_z$$



$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z, \quad \mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

► **Dot product of the two vectors $\mathbf{A}$ and $\mathbf{B}$**

⚡ **Important result**

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

► **Cross product of the two vectors $\mathbf{A}$ and $\mathbf{B}$**

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= A_x B_x \mathbf{a}_x \times \mathbf{a}_x + A_x B_y \mathbf{a}_x \times \mathbf{a}_y + A_x B_z \mathbf{a}_x \times \mathbf{a}_z \\ &\quad + A_y B_x \mathbf{a}_y \times \mathbf{a}_x + A_y B_y \mathbf{a}_y \times \mathbf{a}_y + A_y B_z \mathbf{a}_y \times \mathbf{a}_z \\ &\quad + A_z B_x \mathbf{a}_z \times \mathbf{a}_x + A_z B_y \mathbf{a}_z \times \mathbf{a}_y + A_z B_z \mathbf{a}_z \times \mathbf{a}_z \\ &= (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z \end{aligned}$$

⚡ **Important result**

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

- Differential length → **vector**

$$d\ell = \mathbf{a}_x dx + \mathbf{a}_y dy + \mathbf{a}_z dz$$

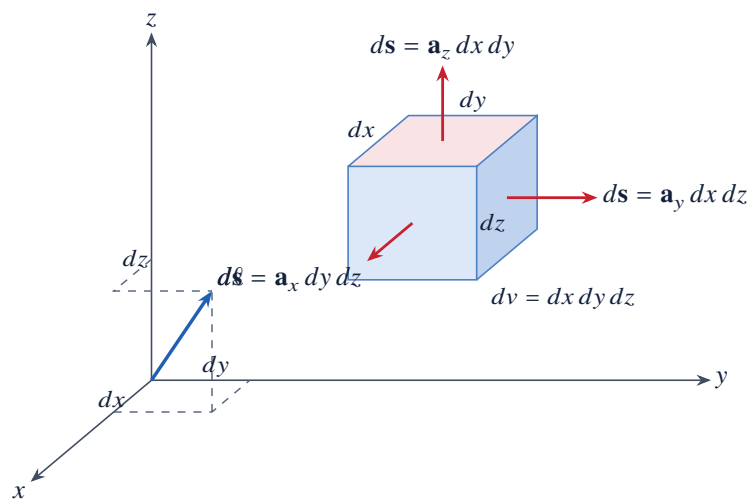
- Differential surface → **vector**

$$d\mathbf{S} = dS \mathbf{a}_n$$

$$d\mathbf{S} = dy dz \mathbf{a}_x, \quad d\mathbf{S} = dx dz \mathbf{a}_y, \quad d\mathbf{S} = dx dy \mathbf{a}_z$$

- Differential volume → **scalar**

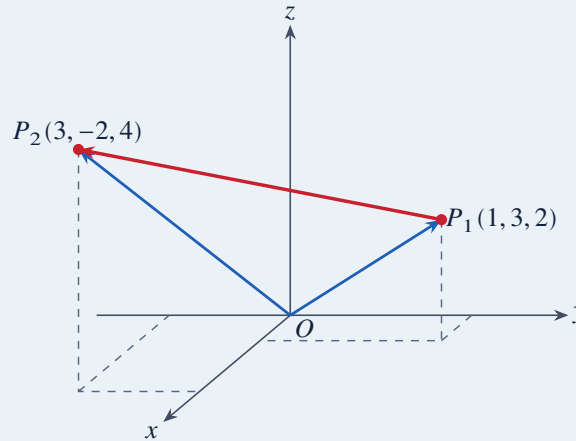
$$dv = dx dy dz$$



Example 1-6

- (a) Find the vector whose initial point is $P_1(1, 3, 2)$ and whose terminal point is $P_2(3, -2, 4)$.
 (b) What is the length of this vector?

► Solution.



$$\begin{aligned}
 \overrightarrow{P_1P_2} &= \overrightarrow{OP_2} - \overrightarrow{OP_1} \\
 (a) \quad &= (\mathbf{a}_x 3 - \mathbf{a}_y 2 + \mathbf{a}_z 4) - (\mathbf{a}_x + \mathbf{a}_y 3 + \mathbf{a}_z 2) \\
 &= \mathbf{a}_x 2 - \mathbf{a}_y 5 + \mathbf{a}_z 2 \\
 (b) \quad P_1P_2 &= |\overrightarrow{P_1P_2}| = \sqrt{2^2 + (-5)^2 + 2^2} = \sqrt{33}
 \end{aligned}$$

Example 1-7

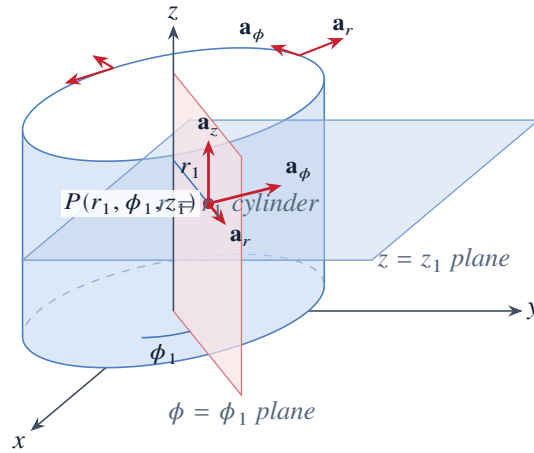
Given $\mathbf{A} = \mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z$, find the unit vector $\mathbf{B}$ such that

- (a) $\mathbf{B} \parallel \mathbf{A}$
- (b) $\mathbf{B}$ lies in the xy -plane and is perpendicular to $\mathbf{A}$

► Solution.

$$\begin{aligned}
 (a) \quad \mathbf{B} &= \frac{\mathbf{A}}{A} = \frac{\mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z}{\sqrt{5^2 + 2^2 + 1^2}} = \frac{\mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z}{\sqrt{30}} \\
 (b) \quad &\begin{cases} B = \sqrt{B_x^2 + B_y^2 + B_z^2} = 1 \\ B_z = 0 \\ \mathbf{B} \cdot \mathbf{A} = 5B_x - 2B_y = 0 \end{cases} \Rightarrow \begin{cases} B_x = \frac{2}{\sqrt{29}} \\ B_y = \frac{5}{\sqrt{29}} \end{cases}
 \end{aligned}$$

1.12 Cylindrical Coordinate System



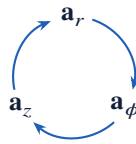
- Every point $P(r_1, \phi_1, z_1)$ is the intersection of the cylinder $r = r_1$, the half-plane $\phi = \phi_1$ and the plane $z = z_1$.
- Unit vector $\mathbf{a}_r \rightarrow$ the unit vector normal to the r -constant cylinder, in the positive direction
- Unit vector $\mathbf{a}_\phi \rightarrow$ the unit vector normal to the ϕ -constant half-plane, in the positive direction
- Unit vector $\mathbf{a}_z \rightarrow$ the unit vector normal to the z -constant plane, in the positive direction

① Remark

The directions of the vectors $\mathbf{a}_r$ and $\mathbf{a}_\phi$ change from point to point in space. This must be taken into account in integration.

- Cross and dot products of the unit vectors

$$\mathbf{a}_r \times \mathbf{a}_\phi = \mathbf{a}_z, \quad \mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_r, \quad \mathbf{a}_z \times \mathbf{a}_r = \mathbf{a}_\phi$$



$$\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_r \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_r = 0$$

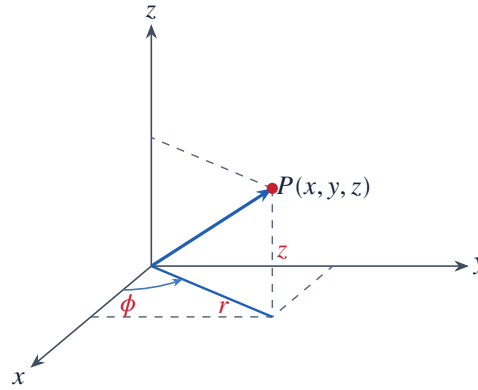
- Representation of an arbitrary vector $\mathbf{A}$ in cylindrical coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$

$$A = \sqrt{A_r^2 + A_\phi^2 + A_z^2}$$

► Position vector

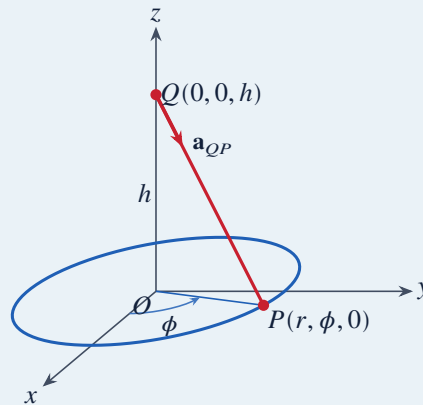


$$\mathbf{R} = r\mathbf{a}_r + z\mathbf{a}_z$$

Example 1-8

The cylindrical coordinates of an arbitrary point P are $(r, \phi, 0)$. Find the unit vector from the point $z = h$ on the z -axis toward the point P .

► Solution.



$$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = (\mathbf{a}_r r) - (\mathbf{a}_z h)$$

$$\mathbf{a}_{QP} = \frac{\overrightarrow{QP}}{|\overrightarrow{QP}|} = \frac{1}{\sqrt{r^2 + h^2}} (\mathbf{a}_r r - \mathbf{a}_z h)$$

► Differential length → vector

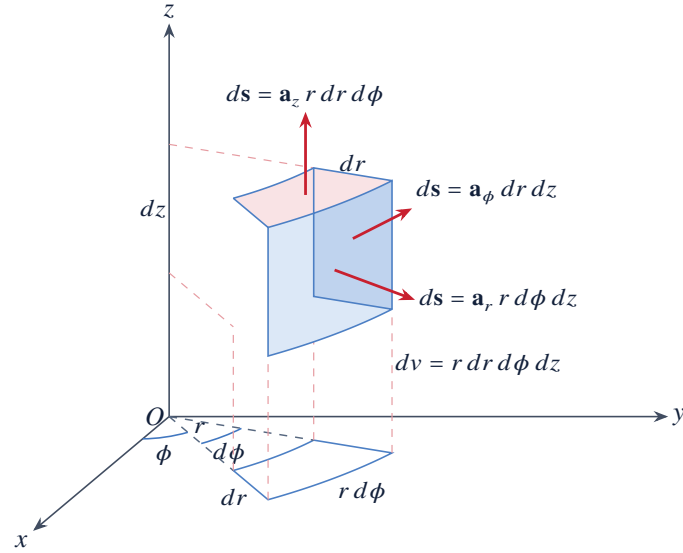
$$d\ell = \mathbf{a}_r dr + \mathbf{a}_\phi r d\phi + \mathbf{a}_z dz$$

► Differential surface → vector

$$d\mathbf{S} = r d\phi dz \mathbf{a}_r, \quad d\mathbf{S} = dr dz \mathbf{a}_\phi, \quad d\mathbf{S} = r dr d\phi \mathbf{a}_z$$

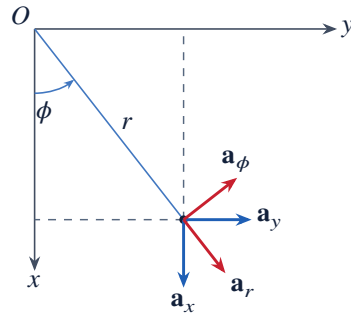
► Differential volume → scalar

$$dv = r dr d\phi dz$$



- Transformation of the representation of a vector from cylindrical to Cartesian coordinates

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$



$$\mathbf{a}_r = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y, \quad \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y, \quad \mathbf{a}_z = \mathbf{a}_z$$

$$\begin{aligned} \mathbf{A} &= (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) A_r + (-\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y) A_\phi + \mathbf{a}_z A_z \\ &\Rightarrow \mathbf{A} = \mathbf{a}_x (\cos \phi A_r - \sin \phi A_\phi) + \mathbf{a}_y (\sin \phi A_r + \cos \phi A_\phi) + \mathbf{a}_z A_z \end{aligned}$$

- Transformation of the representation of a vector from Cartesian to cylindrical coordinates

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$

$$\mathbf{a}_x = \cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi, \quad \mathbf{a}_y = \sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi, \quad \mathbf{a}_z = \mathbf{a}_z$$

$$\begin{aligned} \mathbf{A} &= (\cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi) A_x + (\sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi) A_y + \mathbf{a}_z A_z \\ &\Rightarrow \mathbf{A} = \mathbf{a}_r (\cos \phi A_x + \sin \phi A_y) + \mathbf{a}_\phi (-\sin \phi A_x + \cos \phi A_y) + \mathbf{a}_z A_z \end{aligned}$$

- Transformation of the representation of a vector from cylindrical to Cartesian coordinates

⚡ Important result

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from Cartesian to cylindrical coordinates

⚡ Important result

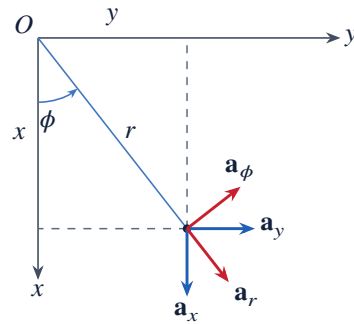
$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Transformation from cylindrical to Cartesian coordinates

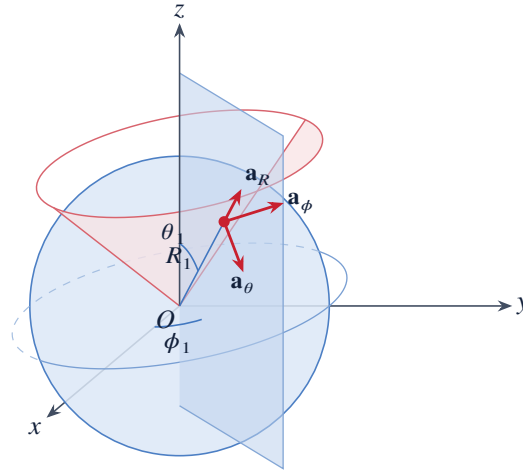
$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z$$

- Transformation from Cartesian to cylindrical coordinates

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$$



1.13 Spherical Coordinate System



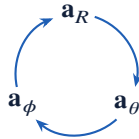
- Every point $P(R_1, \theta_1, \phi_1)$ is the intersection of the sphere $R = R_1$, the cone $\theta = \theta_1$ and the half-plane $\phi = \phi_1$
- Unit vector $\mathbf{a}_R \rightarrow$ the unit vector normal to the R -constant sphere, in the positive direction
- Unit vector $\mathbf{a}_\theta \rightarrow$ the unit vector normal to the θ -constant cone, in the positive direction
- Unit vector $\mathbf{a}_\phi \rightarrow$ the unit vector normal to the ϕ -constant half-plane, in the positive direction

Remark

The directions of the vectors $\mathbf{a}_R$, $\mathbf{a}_\theta$ and $\mathbf{a}_\phi$ change from point to point in space. This must be taken into account in integration.

- Cross and dot products of the unit vectors

$$\mathbf{a}_R \times \mathbf{a}_\theta = \mathbf{a}_\phi, \quad \mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_R, \quad \mathbf{a}_\phi \times \mathbf{a}_R = \mathbf{a}_\theta$$



$$\mathbf{a}_R \cdot \mathbf{a}_R = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

$$\mathbf{a}_R \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_R = 0$$

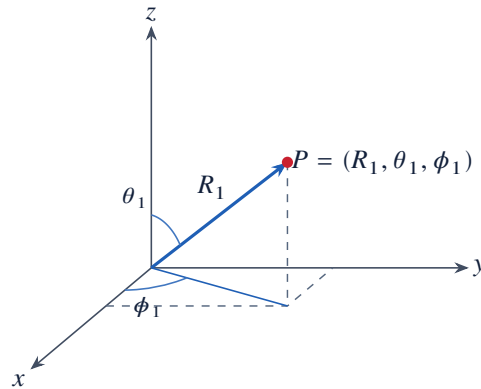
- Representation of an arbitrary vector $\mathbf{A}$ in spherical coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$A = \sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$$

- Position vector



$$\mathbf{R} = R_1 \mathbf{a}_R$$

- Differential length → **vector**

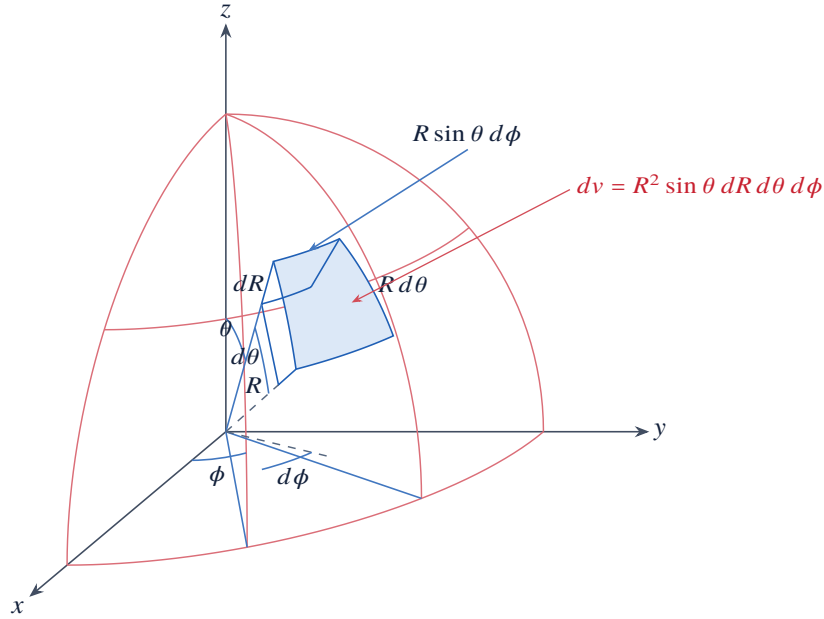
$$d\ell = \mathbf{a}_R dR + \mathbf{a}_\theta R d\theta + \mathbf{a}_\phi R \sin \theta d\phi$$

- Differential surface → **vector**

$$d\mathbf{S} = R^2 \sin \theta d\theta d\phi \mathbf{a}_R, \quad d\mathbf{S} = R \sin \theta dR d\phi \mathbf{a}_\theta, \quad d\mathbf{S} = R dR d\theta \mathbf{a}_\phi$$

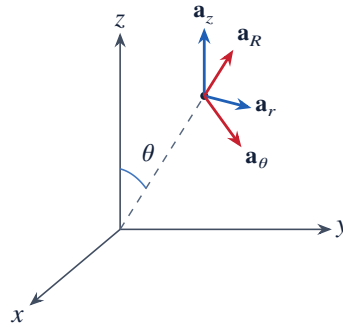
- Differential volume → **scalar**

$$dv = R^2 \sin \theta dR d\theta d\phi$$



- Transformation of the representation of a vector from spherical to cylindrical coordinates

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$



$$\mathbf{a}_R = \sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z, \quad \mathbf{a}_\theta = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z, \quad \mathbf{a}_\phi = \mathbf{a}_\phi$$

$$\begin{aligned} \mathbf{A} &= (\sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z) A_R + (\cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z) A_\theta + \mathbf{a}_\phi A_\phi \\ &\Rightarrow \mathbf{A} = \mathbf{a}_r (\sin \theta A_R + \cos \theta A_\theta) + \mathbf{a}_\phi A_\phi + \mathbf{a}_z (\cos \theta A_R - \sin \theta A_\theta) \end{aligned}$$

- Transformation of the representation of a vector from cylindrical to spherical coordinates

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$\mathbf{a}_r = \sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta, \quad \mathbf{a}_\phi = \mathbf{a}_\phi, \quad \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta$$

$$\begin{aligned} \mathbf{A} &= (\sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta) A_r + \mathbf{a}_\phi A_\phi + (\cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta) A_z \\ &\Rightarrow \mathbf{A} = \mathbf{a}_R (\sin \theta A_r + \cos \theta A_z) + \mathbf{a}_\theta (\cos \theta A_r - \sin \theta A_z) + \mathbf{a}_\phi A_\phi \end{aligned}$$

- Transformation of the representation of a vector from spherical to cylindrical coordinates

⚡ Important result

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix}$$

- Transformation of the representation of a vector from cylindrical to spherical coordinates

⚡ Important result

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from Cartesian to spherical coordinates

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$\begin{cases} \mathbf{a}_x = \cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y = \sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z = \mathbf{a}_z \end{cases} \quad \begin{cases} \mathbf{a}_r = \sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta \\ \mathbf{a}_\phi = \mathbf{a}_\phi \\ \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta \end{cases}$$

$$\Rightarrow \begin{cases} \mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_R + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_R + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta \end{cases}$$

⚡ Important result

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from spherical to Cartesian coordinates

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$

$$\begin{cases} \mathbf{a}_R = \sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z \\ \mathbf{a}_\theta = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z \\ \mathbf{a}_\phi = \mathbf{a}_\phi \end{cases} \quad \begin{cases} \mathbf{a}_r = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y \\ \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \\ \mathbf{a}_z = \mathbf{a}_z \end{cases}$$

$$\Rightarrow \begin{cases} \mathbf{a}_R = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z \\ \mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z \\ \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \end{cases}$$

⚡ Important result

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix}$$

- Transformation from cylindrical to spherical coordinates

$$R = \sqrt{r^2 + z^2}, \quad \theta = \tan^{-1} \frac{r}{z}, \quad \phi = \phi$$

- Transformation from spherical to cylindrical coordinates

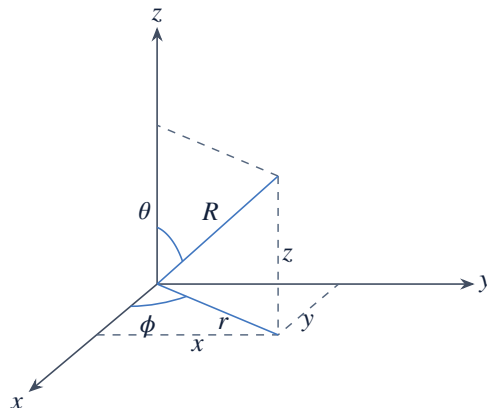
$$r = R \sin \theta, \quad \phi = \phi, \quad z = R \cos \theta$$

- Transformation from spherical to Cartesian coordinates

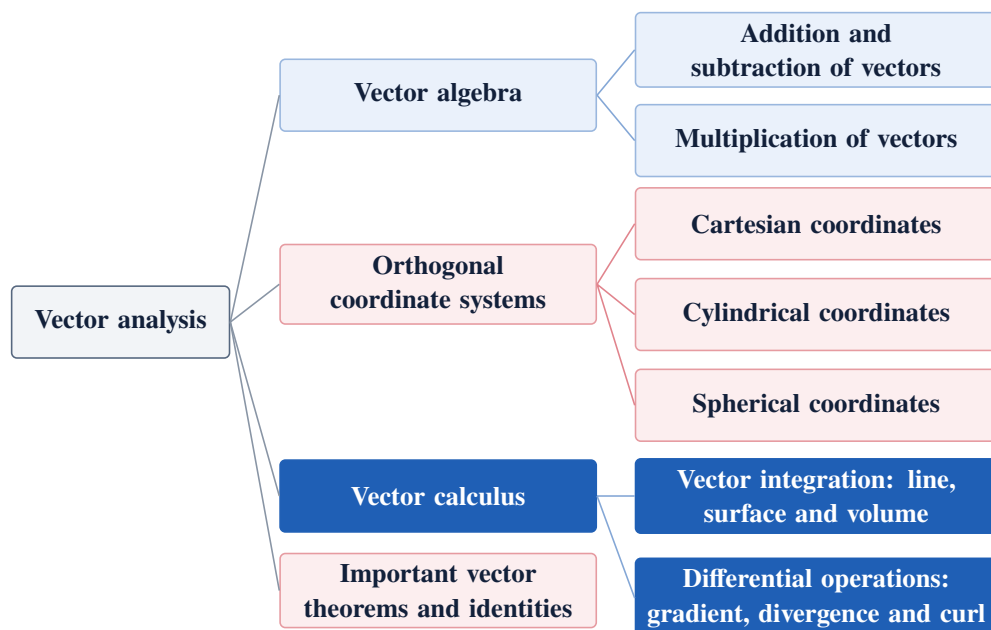
$$x = R \sin \theta \cos \phi, \quad y = R \sin \theta \sin \phi, \quad z = R \cos \theta$$

- Transformation from Cartesian to spherical coordinates

$$R = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x}$$



1.14 Integrals Containing Vector Functions



- The kinds of integrals containing vector functions that we deal with

$$\begin{aligned}
 &\int_C \mathbf{F} d\ell \quad \iint_S \mathbf{F} ds \quad \iiint_V \mathbf{F} dv \\
 &\int_C V d\ell \\
 &\int_C \mathbf{F} \cdot d\ell \quad \iint_S \mathbf{A} \cdot ds \\
 &\int_C \mathbf{F} \times d\ell
 \end{aligned}$$

Remark

Important points about integration involving vector functions

- Using an appropriate coordinate system
- Correct determination of the differential length, surface and volume
- Attention to the variation of the unit vectors $\mathbf{a}_r$, $\mathbf{a}_R$, $\mathbf{a}_\theta$ and $\mathbf{a}_\phi$ over the range of integration

- The integral of a vector function over a specified curve, surface and volume

$$\int_C \mathbf{F} d\ell \quad \iint_S \mathbf{F} ds \quad \iiint_V \mathbf{F} dv$$

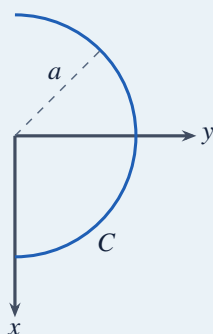
- The result is a vector.
- Important applications in electromagnetics: computing the electric field intensity due to a line, surface and volume charge distribution

$$\begin{cases} \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int_C \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ \mathbf{E} = \frac{1}{4\pi\epsilon_0} \iint_S \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ \mathbf{E} = \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho_v dv' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \end{cases}$$

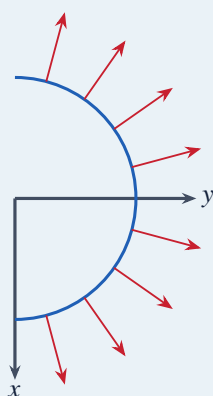
Example 1-9

Evaluate the following integral over the curve C .

$$\int_C \mathbf{F} d\ell, \quad \mathbf{F} = 3a\mathbf{a}_r$$



► **Solution.**



$$d\ell = r d\phi = a d\phi$$

$$\int_C \mathbf{F} d\ell = \int_0^\pi 3a\mathbf{a}_r a d\phi = 3a \int_0^\pi (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) d\phi = 6a\mathbf{a}_y$$

- ▶ The integral of a scalar function along a specified path

$$\int_C V \, d\ell$$

- ▶ The result is a vector.
- ▶ Important applications in electromagnetics: computing the magnetic potential due to a line current

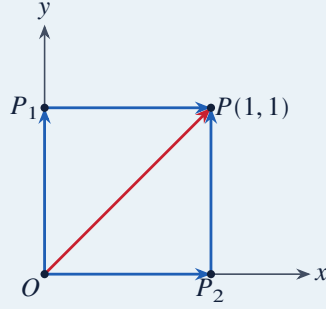
$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

Example 1-10

Evaluate the following integral along the following paths

$$\int_O^P r^2 d\ell, \quad r^2 = x^2 + y^2$$

- Path OP
- Path OP_1P
- Path OP_2P



➤ Solution.

- Along the path OP

$$\begin{aligned} \int_O^P r^2 d\ell &= \mathbf{a}_r \int_0^{\sqrt{2}} r^2 dr = \mathbf{a}_r \frac{2\sqrt{2}}{3} \\ &= \frac{2\sqrt{2}}{3} (\mathbf{a}_x \cos 45^\circ + \mathbf{a}_y \sin 45^\circ) \\ &= \mathbf{a}_x \frac{2}{3} + \mathbf{a}_y \frac{2}{3} \end{aligned}$$

- Along the path OP_1P

$$\begin{aligned} \int_O^P (x^2 + y^2) d\ell &= \mathbf{a}_y \int_O^{P_1} y^2 dy + \mathbf{a}_x \int_{P_1}^P (x^2 + 1) dx \\ &= \mathbf{a}_y \left. \frac{1}{3} y^3 \right|_0^1 + \mathbf{a}_x \left(\left. \frac{1}{3} x^3 + x \right|_0^1 \right) \\ &= \mathbf{a}_x \frac{4}{3} + \mathbf{a}_y \frac{1}{3} \end{aligned}$$

- Along the path OP_2P

$$\begin{aligned} \int_O^P (x^2 + y^2) d\ell &= \mathbf{a}_x \int_O^{P_2} x^2 dx + \mathbf{a}_y \int_{P_2}^P (1 + y^2) dy \\ &= \mathbf{a}_x \left. \frac{1}{3} x^3 \right|_0^1 + \mathbf{a}_y \left(\left. y + \frac{1}{3} y^3 \right|_0^1 \right) \\ &= \mathbf{a}_x \frac{1}{3} + \mathbf{a}_y \frac{4}{3} \end{aligned}$$

► Scalar line integral

$$\int_C \mathbf{F} \cdot d\ell$$

► The result is a scalar.

► If $\mathbf{F}$ is a force

- the work done by the force in moving a body from point P_1 to point P_2 along the specified path C

► Important applications in electromagnetics

- Computing the electric potential difference between the points P_1 and P_2 :

$$V = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

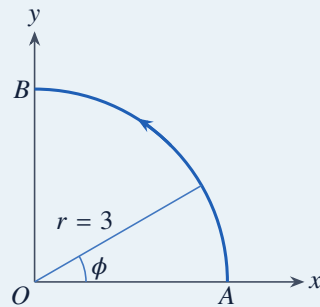
- Ampère's circuital law:

$$\mu_0 I = - \oint_C \mathbf{B} \cdot d\ell$$

Example 1-11

Evaluate the following integral along the quarter-circle path of the figure.

$$\int_A^B \mathbf{F} \cdot d\ell, \quad \mathbf{F} = \mathbf{a}_x xy - \mathbf{a}_y 2x$$



► Solution.

$$\begin{bmatrix} F_r \\ F_\phi \\ F_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} xy \\ -2x \\ 0 \end{bmatrix}$$

$$\mathbf{F} = \mathbf{a}_r(xy \cos \phi - 2x \sin \phi) - \mathbf{a}_\phi(xy \sin \phi + 2x \cos \phi)$$

$$d\ell = \mathbf{a}_\phi 3 d\phi \quad \mathbf{F} \cdot d\ell = -3(xy \sin \phi + 2x \cos \phi) d\phi$$

$$x = 3 \cos \phi \quad y = 3 \sin \phi$$

$$\int_A^B \mathbf{F} \cdot d\ell = \int_0^{\pi/2} -3(9 \sin^2 \phi \cos \phi + 6 \cos^2 \phi) d\phi = -9 \left(1 + \frac{\pi}{2} \right)$$

- Scalar surface integral

$$\iint_S \mathbf{A} \cdot d\mathbf{s}$$

- The result is a scalar.
- Result of the integral $\rightarrow$ the flux of the vector field $\mathbf{A}$ passing through the surface S
- Vector differential surface

$$d\mathbf{s} = \mathbf{a}_n ds$$

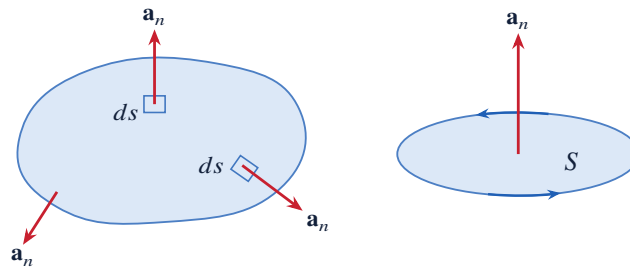


Figure 1.2 | Direction of $\mathbf{a}_n$ for a closed surface: normal to the surface, pointing out of the volume; for an open surface: depends on the direction of travel along the rim of the open surface, by the right-hand rule

- Important applications in electromagnetics

- Gauss's law in free space:

$$\oiint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

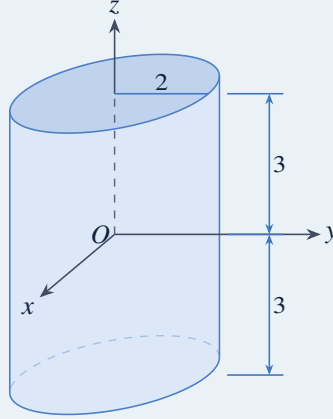
- Computing the electric current:

$$\iint_S \mathbf{J} \cdot d\mathbf{s} = I$$

Example 1-12

Evaluate the following integral over the closed surface about the z -axis specified by $z = \pm 3$ and $r = 2$

$$\oint_S \mathbf{F} \cdot d\mathbf{s}, \quad \mathbf{F} = \mathbf{a}_r \frac{k_1}{r} + \mathbf{a}_z k_2 z$$



► Solution.

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = \oint_S \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} = \int_{\text{top face}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} + \int_{\text{bottom face}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} + \int_{\text{side wall}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s}$$

► Top face

$$z = 3, \quad \mathbf{a}_n = \mathbf{a}_z, \quad \mathbf{F} \cdot \mathbf{a}_n = k_2 z = 3k_2, \quad d\mathbf{s} = r dr d\phi$$

$$\int_{\text{top face}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} = \int_0^{2\pi} \int_0^2 3k_2 r dr d\phi = 12\pi k_2$$

► Bottom face

$$z = -3, \quad \mathbf{a}_n = -\mathbf{a}_z, \quad \mathbf{F} \cdot \mathbf{a}_n = -k_2 z = 3k_2, \quad d\mathbf{s} = r dr d\phi$$

$$\int_{\text{bottom face}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} = 12\pi k_2$$

► Side wall

$$r = 2, \quad \mathbf{a}_n = \mathbf{a}_r, \quad \mathbf{F} \cdot \mathbf{a}_n = \frac{k_1}{r} = \frac{k_1}{2}, \quad d\mathbf{s} = r d\phi dz = 2 d\phi dz$$

$$\int_{\text{side wall}} \mathbf{F} \cdot \mathbf{a}_n d\mathbf{s} = \int_{-3}^3 \int_0^{2\pi} k_1 d\phi dz = 12\pi k_1$$

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = 12\pi k_2 + 12\pi k_2 + 12\pi k_1 = 12\pi(k_1 + 2k_2)$$

► Vector line integral

$$\int_C \mathbf{F} \times d\ell$$

- The result is a vector.
- Important applications in electromagnetics

– Biot–Savart law:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\ell \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3}$$

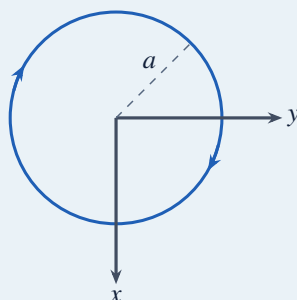
– Computing the magnetic force on a current-carrying wire in a magnetic field:

$$\mathbf{F} = I \oint_C d\ell \times \mathbf{B}$$

Example 1-13

Evaluate the following integral along the path C .

$$\int_C \mathbf{F} \times d\ell, \quad \mathbf{F} = 2\mathbf{a}_z$$



► Solution.

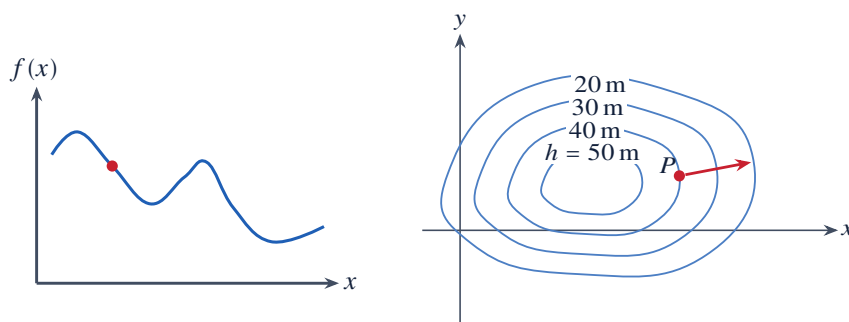
$$d\ell = r d\phi \mathbf{a}_\phi = a d\phi \mathbf{a}_\phi$$

$$\begin{aligned} \int_C \mathbf{F} \times d\ell &= \int_{2\pi}^0 2\mathbf{a}_z \times a d\phi \mathbf{a}_\phi = -2a \int_{2\pi}^0 \mathbf{a}_r d\phi \\ &= -2a \int_{2\pi}^0 (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) d\phi = 0 \end{aligned}$$

1.15 Differential Operations - Gradient

► Prerequisite concepts for the discussion of the gradient

- The concept of the derivative of a function of one variable
 - The rate of change of the function at the point of interest
- The concept of the directional derivative of a function of several variables
 - The rate of change of the function at the point of interest and in a specified direction



△ Definition | Gradient

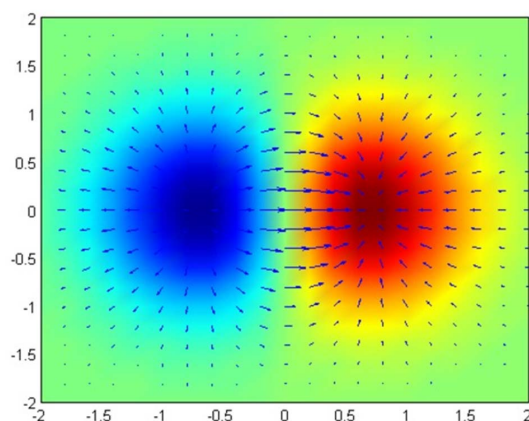
The gradient of a **scalar quantity** is a **vector quantity** that lies in the direction in which the scalar quantity increases most, and whose magnitude equals the maximum rate of increase of the scalar quantity.

$$\text{grad } v = \nabla v = \mathbf{a}_n \frac{dv}{dn}$$

- The directional derivative of v in the direction of $d\ell$:

$$\frac{dv}{d\ell} = \nabla v \cdot \mathbf{a}_\ell$$

- The gradient of v is normal to the surfaces of constant v .



- The gradient in Cartesian coordinates is as follows

⚡ Important result

$$\nabla v = \frac{\partial v}{\partial x} \mathbf{a}_x + \frac{\partial v}{\partial y} \mathbf{a}_y + \frac{\partial v}{\partial z} \mathbf{a}_z$$

- The general expression of the gradient

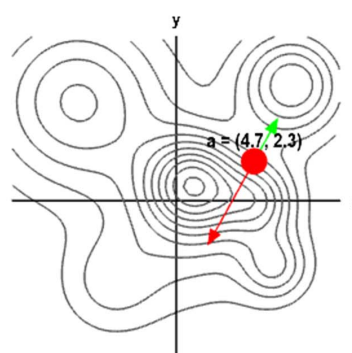
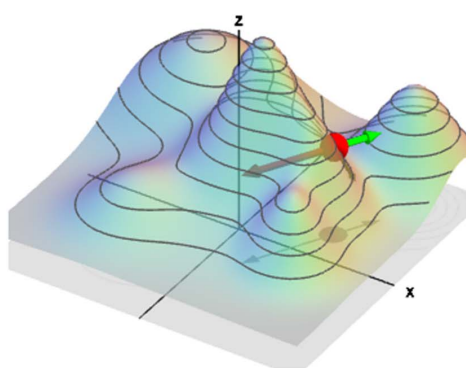
⚡ Important result

$$\nabla V = \mathbf{a}_{u_1} \frac{\partial V}{h_1 \partial u_1} + \mathbf{a}_{u_2} \frac{\partial V}{h_2 \partial u_2} + \mathbf{a}_{u_3} \frac{\partial V}{h_3 \partial u_3}$$

Metric coefficients	Cartesian	Cylindrical	Spherical
h_1	1	1	1
h_2	1	r	R
h_3	1	1	$R \sin \theta$

- **The concept of the gradient**

- Representing a hill by contours of height $\rightarrow h(x, y)$
- If a ball is placed at point P , in which direction does it move and what is the magnitude of the force acting on it?
 - The answer is proportional to $-\nabla h$.



Example 1-14

The electric field intensity $\mathbf{E}$ can be obtained as the negative of the gradient of the scalar electric potential V . Find $\mathbf{E}$ at the point $(1, 1, 0)$ if

$$V = E_0 R \cos \theta$$

► **Solution.**

$$\mathbf{E} = -\nabla V$$

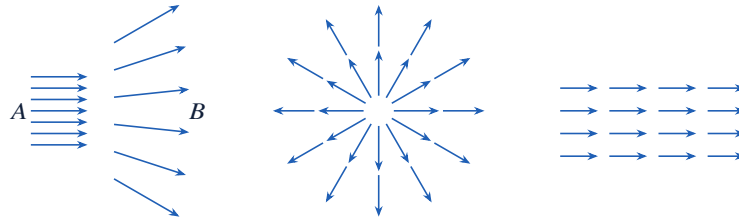
$$\begin{aligned} \mathbf{E} &= - \left[\mathbf{a}_R \frac{\partial}{\partial R} + \mathbf{a}_\theta \frac{1}{R} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi} \right] E_0 R \cos \theta \\ &= -(\mathbf{a}_R \cos \theta - \mathbf{a}_\theta \sin \theta) E_0 \end{aligned}$$

$$\mathbf{E} = -\mathbf{a}_z E_0$$

1.16 Differential Operations - Divergence

► Prerequisite concepts for the discussion of the divergence

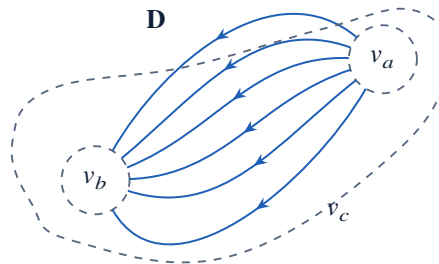
- In studying vector fields it is easier to represent the variations of the field graphically by flux lines. These are directed lines or curves that indicate the direction of the vector field at each point.



- The flux of a vector field is analogous to the flow of a fluid such as water.
- If $\mathbf{A}$ is a flux density vector, the total flux leaving the surface S equals

$$\oint_S \mathbf{A} \cdot d\mathbf{s}$$

- In a volume with a closed surface, there will be an excess of inward or outward flux only when the volume contains a sink or a source, respectively.
 - The net outward flux from v_a is positive: there is a source in the volume v_a
 - The net outward flux from v_b is negative: there is a sink in the volume v_b
 - The net outward flux from v_c is zero.



≡ Definition | Divergence

The divergence of a **vector field** $\mathbf{A}$ at a point is the net outward flux of $\mathbf{A}$ per unit volume as the volume about the point tends to zero

$$\text{div } \mathbf{A} \triangleq \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{s}}{\Delta v}$$

- In Cartesian coordinates the divergence is as follows

⚡ Important result

$$\operatorname{div} \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad \nabla \cdot \mathbf{A} \equiv \operatorname{div} \mathbf{A}$$

- The general expression of the divergence

⚡ Important result

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 A_1) + \frac{\partial}{\partial u_2} (h_1 h_3 A_2) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right]$$

🔗 Example 1-15

The magnetic flux density $\mathbf{B}$ outside a long current-carrying wire is circular and inversely proportional to the distance from the axis of the wire. Find the divergence of $\mathbf{B}$.

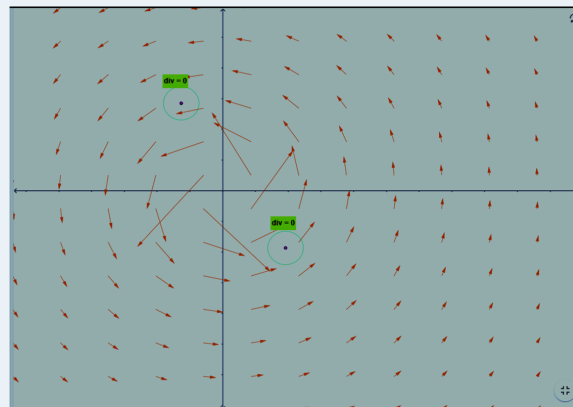
► Solution.

- Assuming the wire coincides with the z -axis

$$\mathbf{B} = \mathbf{a}_\phi \frac{k}{r}$$

$$\nabla \cdot \mathbf{B} = \frac{1}{r} \frac{\partial}{\partial r} (r B_r) + \frac{1}{r} \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z}$$

$$\nabla \cdot \mathbf{B} = 0$$



1.17 Differential Operations - Curl

► Prerequisite concepts for the discussion of the curl

- The circulation of a vector field around a closed path

$$\text{Circulation of } \mathbf{A} \text{ around contour } C \triangleq \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

- The physical meaning of the circulation depends on the kind of field that the vector $\mathbf{A}$ represents.
 - If $\mathbf{A}$ is the force acting on a body, the circulation of $\mathbf{A}$ is the work done by the force in moving the body around the path.

$\triangleq$ Definition | Curl

The curl of a **vector field** $\mathbf{A}$ at a point is a **vector** whose magnitude is the maximum net circulation of $\mathbf{A}$ per unit area as the area about the point tends to zero, and whose direction is the direction of the normal to the area when the area is oriented so as to make the net circulation maximum.

$$\text{curl } \mathbf{A} \equiv \nabla \times \mathbf{A} \triangleq \lim_{\Delta s \rightarrow 0} \frac{1}{\Delta s} \left[\mathbf{a}_n \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \right]_{\max}$$

- In Cartesian coordinates the curl is as follows

⚡ Important result

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

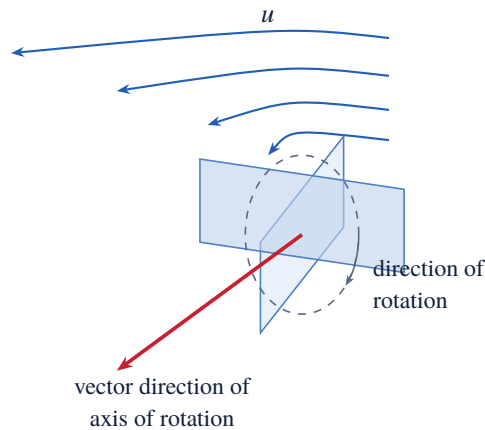
- The general expression of the curl

⚡ Important result

$$\nabla \times \mathbf{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \mathbf{a}_{u_1} h_1 & \mathbf{a}_{u_2} h_2 & \mathbf{a}_{u_3} h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

► **The concept of the curl**

- The rotation of a paddle wheel in a stream of water with velocity vector u
 - The maximum angular velocity of the paddle wheel is proportional to the magnitude of the curl of the velocity vector.
 - The axis of the wheel, by the right-hand rule, lies in the direction of the curl.



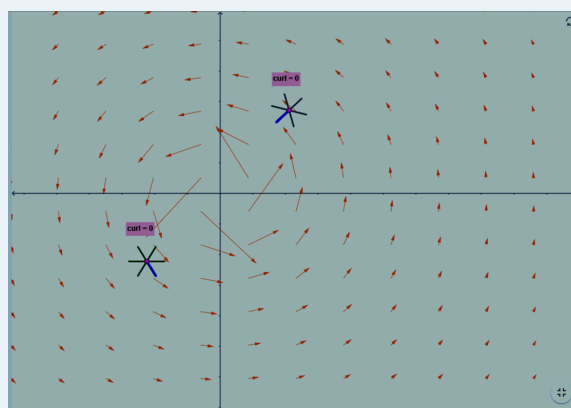
Example 1-16

Find the curl of the following vector

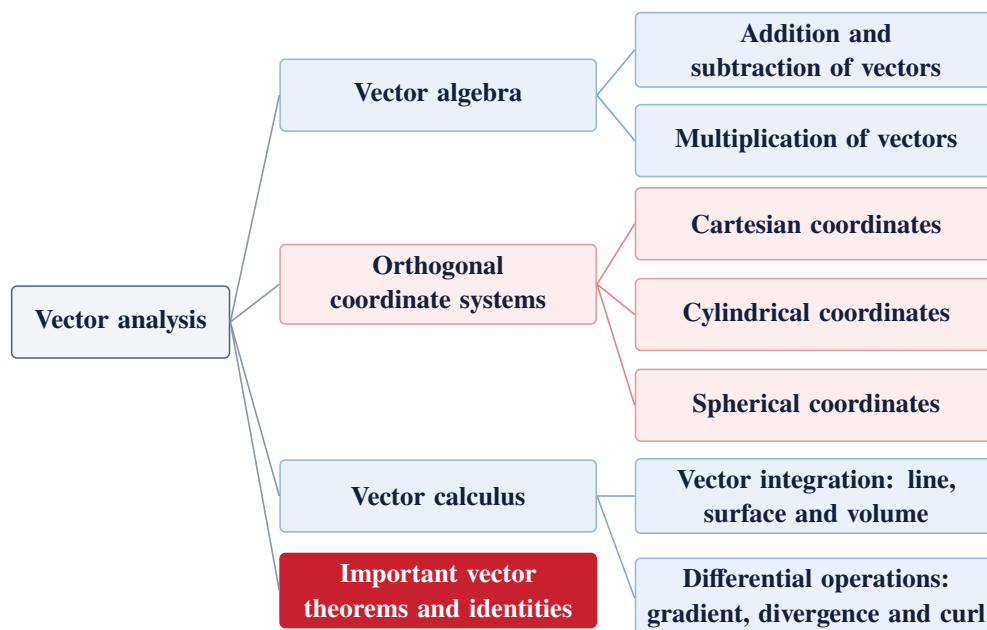
$$\mathbf{A} = \mathbf{a}_\phi \left(\frac{k}{r} \right)$$

► **Solution.**

$$\nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\phi r & \mathbf{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & k & 0 \end{vmatrix} = 0$$



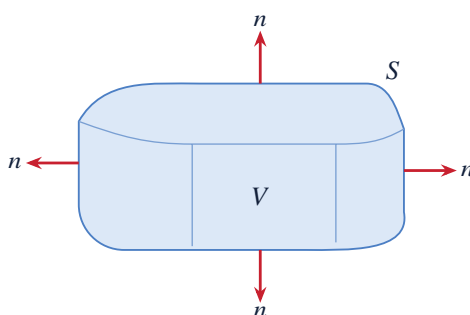
1.18 Important Vector Theorems



▽ Theorem | Divergence theorem

The volume integral of the divergence of a vector field equals the total outward flux of the vector through the surface enclosing the volume.

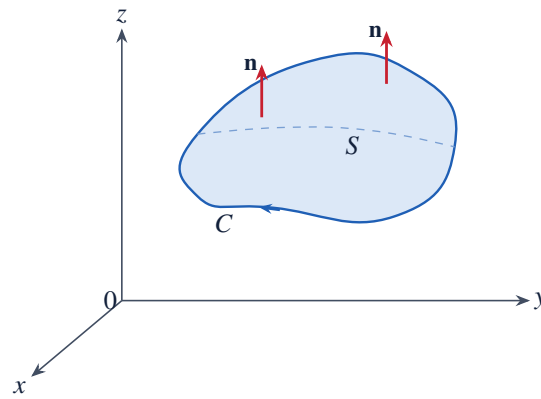
$$\int_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{s}$$



▽ Theorem | Stokes's theorem

The surface integral of the curl of a vector field over an open surface equals the closed line integral of the vector along the contour bounding the surface.

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

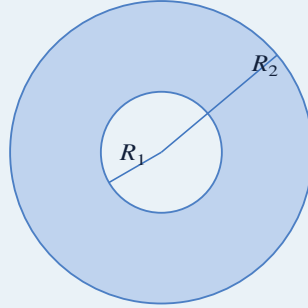
**▽ Theorem** | Helmholtz's theorem

A vector field is determined to within an additive constant if both its divergence and its curl are specified.

Example 1-17

Verify the divergence theorem for the shell region enclosed by the spherical surfaces $R = R_1$ and $R = R_2$ ($R_2 > R_1$) centred at the origin, for the following vector field.

$$\mathbf{F} = \mathbf{a}_R kR$$



► **Solution.**

$$\int_V \nabla \cdot \mathbf{A} dv = \oint_S \mathbf{A} \cdot d\mathbf{s}$$

► For the outer surface

$$R = R_2, \quad d\mathbf{s} = \mathbf{a}_R R_2^2 \sin \theta d\theta d\phi$$

$$\int_{\text{outer surface}} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^\pi (kR_2) R_2^2 \sin \theta d\theta d\phi = 4\pi k R_2^3$$

► For the inner surface

$$R = R_1, \quad d\mathbf{s} = -\mathbf{a}_R R_1^2 \sin \theta d\theta d\phi$$

$$\int_{\text{inner surface}} \mathbf{F} \cdot d\mathbf{s} = - \int_0^{2\pi} \int_0^\pi (kR_1) R_1^2 \sin \theta d\theta d\phi = -4\pi k R_1^3$$

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = 4\pi k (R_2^3 - R_1^3)$$

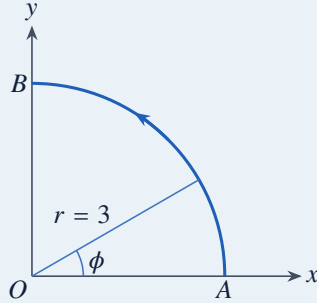
$$\nabla \cdot \mathbf{F} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 F_R) = \frac{1}{R^2} \frac{\partial}{\partial R} (kR^3) = 3k$$

$$\int_V \nabla \cdot \mathbf{F} dv = (\nabla \cdot \mathbf{F}) V = 4\pi k (R_2^3 - R_1^3)$$

Example 1-18

Verify Stokes's theorem over a quarter of a circular disk of radius 3 in the first quadrant for the following vector field.

$$\mathbf{F} = \mathbf{a}_x xy - \mathbf{a}_y 2x$$



► **Solution.**

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -2x & 0 \end{vmatrix} = -\mathbf{a}_z (2 + x)$$

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{s} = \int_0^{\pi/2} \int_0^3 -\mathbf{a}_z (2 + r \cos \phi) \cdot r dr d\phi \mathbf{a}_z = -9 \left(1 + \frac{\pi}{2} \right)$$

► From B to O

$$x = 0, \text{ and } \mathbf{F} \cdot d\boldsymbol{\ell} = \mathbf{F} \cdot (\mathbf{a}_y dy) = 2x dy = 0$$

► From O to A

$$y = 0, \text{ and } \mathbf{F} \cdot d\boldsymbol{\ell} = \mathbf{F} \cdot (\mathbf{a}_x dx) = xy dx = 0$$

$$\oint_{ABOA} \mathbf{F} \cdot d\boldsymbol{\ell} = \int_A^B \mathbf{F} \cdot d\boldsymbol{\ell} = -9 \left(1 + \frac{\pi}{2} \right)$$

(examined in Example 1-11)

1.19 Important Vector Identities

⚡ Important result | Identity I

The curl of the gradient of any scalar quantity is zero

$$\nabla \times (\nabla V) \equiv 0$$

- A curl-free vector field is called an **irrotational** or **conservative** field.
- If a vector quantity is curl-free, it can be expressed as the gradient of a scalar quantity.

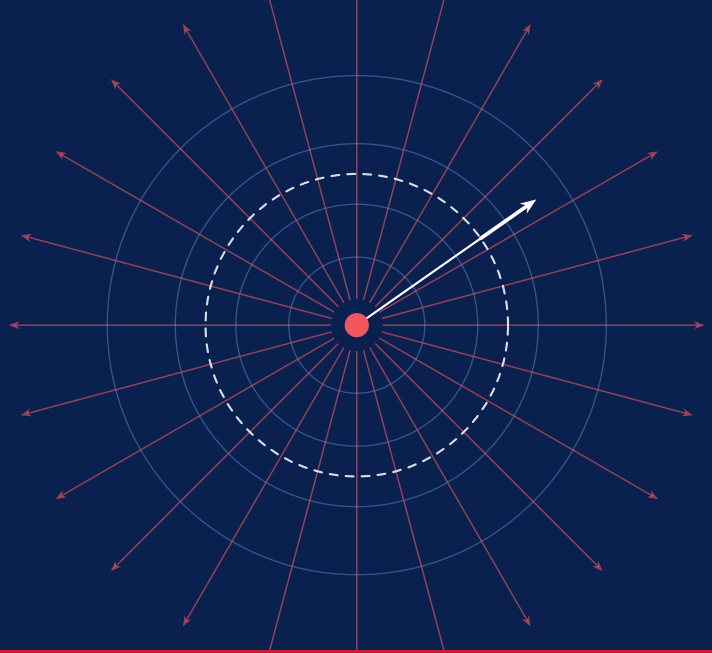
⚡ **Important result** | Identity II

The divergence of the curl of any vector field is zero.

$$\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$$

- A divergence-free vector field is called a **solenoidal** field.
- If a vector quantity is divergence-free, it can be expressed as the curl of a vector quantity.

2



CHAPTER

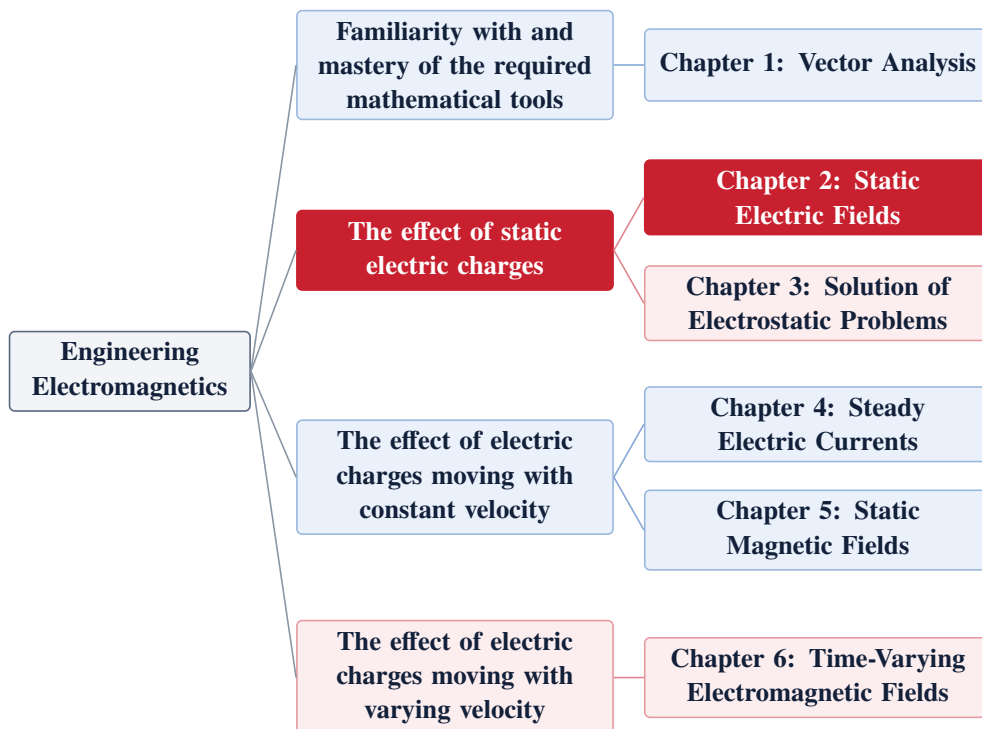
Static Electric Fields

میدان‌های الکتریکی ساکن

In this chapter

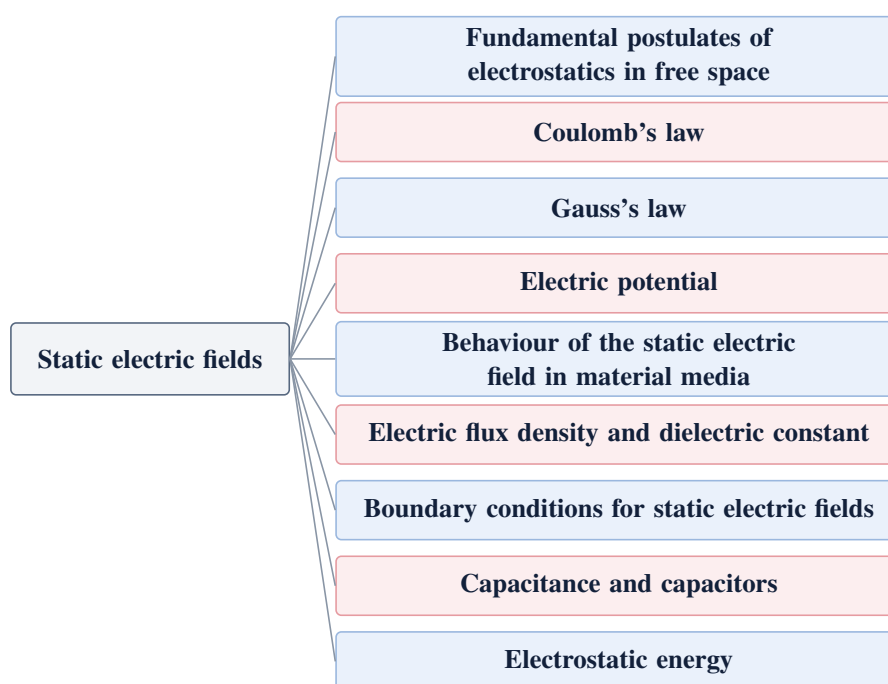
2.1	Course Outline	45	2.8	Conductors in Static Electric Field	69
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2.1 Course Outline



2.2 Static Electric Fields

- In electrostatics the electric charges are at rest and the electric fields do not change with time.
 - In this case there are no magnetic fields.
- In introductory physics the development of electrostatics usually begins with Coulomb's experimental law for the force between two charges.
- In this course, instead of following the historical development of electrostatics, we introduce the subject by postulating the divergence and the curl of the electric field intensity in free space.



- What are the fundamental postulates of electrostatics from which the other relations and laws of the subject can be derived?
- How are Coulomb's and Gauss's laws derived from these postulates?
- How is the electric field of a point charge, of a number of discrete point charges and of a continuous charge distribution computed?

2.3 Fundamental Postulates of Electrostatics in Free Space

► Electric field intensity

- The force on a small test charge placed in the field, divided by the charge. $\rightarrow$ N/C (newtons per coulomb), equivalent to V/m (volts per metre)

$\triangleq$ Definition

$$\mathbf{E} = \lim_{q \rightarrow 0} \frac{\mathbf{F}}{q} \quad (\text{V/m})$$

- The electric field intensity is proportional to the force and in the same direction.
- The two fundamental postulates of electrostatics in free space

⚡ Important result

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \mathbf{E} = 0$$

- ρ : volume charge density (C/m^3); ϵ_0 : permittivity of free space (F/m)
- $\mathbf{E}$ is not solenoidal unless $\rho = 0$
- $\mathbf{E}$ is irrotational
- Integral form of the first equation $\rightarrow$ integration over an arbitrary volume V

$$\int_V \nabla \cdot \mathbf{E} \, dv = \frac{1}{\epsilon_0} \int_V \rho \, dv$$

- $\int_V \rho \, dv$: the total charge contained in the volume V enclosed by the surface S
- Using the divergence theorem:

⚡ Important result

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

- This relation is a form of **Gauss's law**
 - The total outward flux of the electric field intensity through any closed surface equals the total charge enclosed in the surface divided by ϵ_0 .
- Integral form of the second equation $\rightarrow$ integration over an arbitrary open surface S

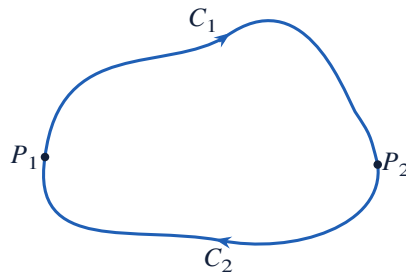
$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = 0$$

- Using Stokes's theorem:

⚡ Important result

$$\oint_C \mathbf{E} \cdot d\ell = 0$$

- The scalar line integral of the static electric field intensity around any closed path is zero.
- The scalar product $\mathbf{E} \cdot d\ell$ integrated over any path is the voltage along that path.
- This equation is an expression of **Kirchhoff's voltage law**
 - The algebraic sum of the voltage drops around any closed path is zero.
- The line integrals of the irrotational field $\mathbf{E}$ are independent of the path and depend only on the initial and final points.



$$\oint_C \mathbf{E} \cdot d\ell = 0$$

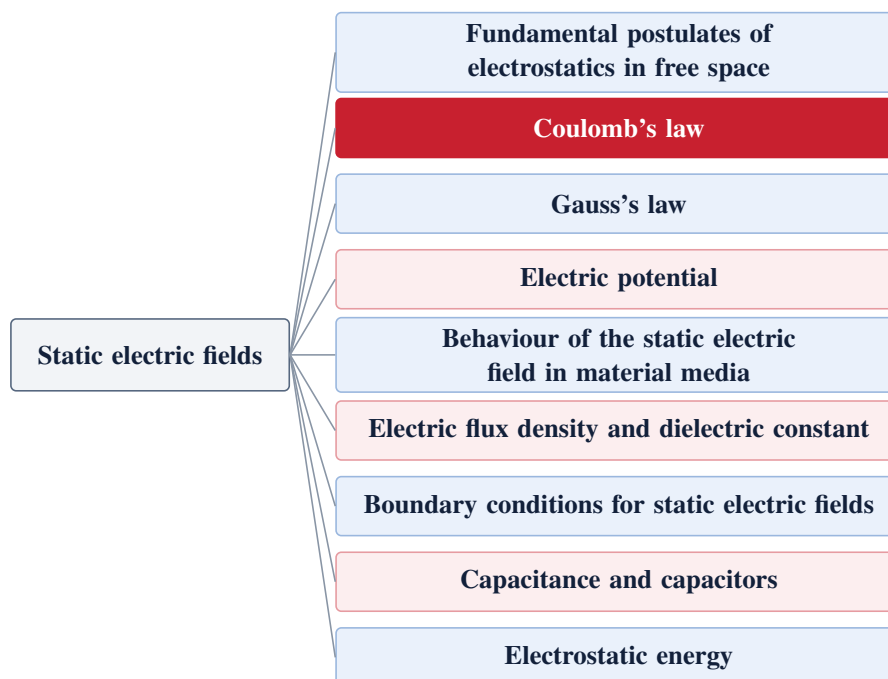
$$\int_{C_1} \mathbf{E} \cdot d\ell + \int_{C_2} \mathbf{E} \cdot d\ell = 0$$

$$\int_{P_1}^{P_2} \mathbf{E} \cdot d\ell + \int_{P_2}^{P_1} \mathbf{E} \cdot d\ell = 0$$

$$\int_{\text{Along } C_1}^{P_2}_{P_1} \mathbf{E} \cdot d\ell = - \int_{\text{Along } C_2}^{P_1}_{P_2} \mathbf{E} \cdot d\ell$$

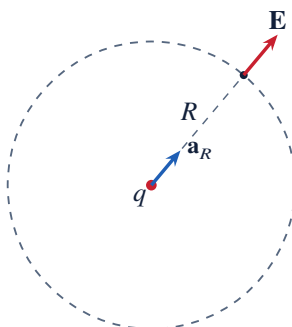
$$\int_{\text{Along } C_1}^{P_2}_{P_1} \mathbf{E} \cdot d\ell = \int_{\text{Along } C_2}^{P_2}_{P_1} \mathbf{E} \cdot d\ell$$

2.4 Coulomb's Law



► The electric field intensity of a point charge q in free space

- We consider a hypothetical spherical surface of radius R centred at q .
- Since a point charge has no preferred direction, we expect the field to be radial everywhere and its intensity to be constant over the spherical surface.



► The electric field intensity on the spherical surface:

$$\mathbf{E} = E_R \mathbf{a}_R$$

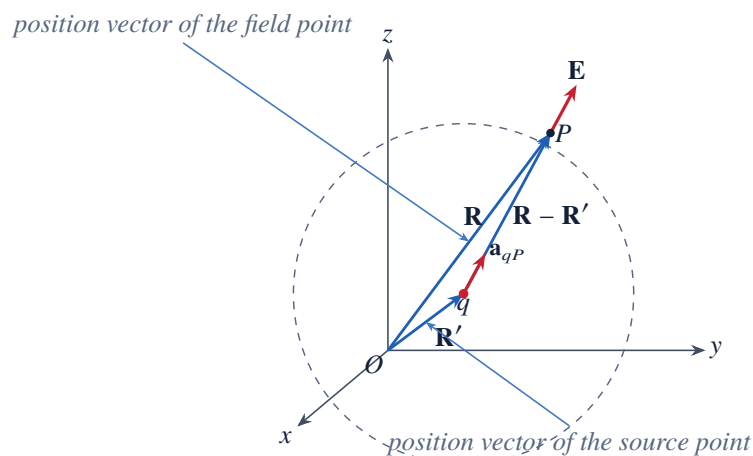
$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \oint_S (\mathbf{a}_R E_R) \cdot \mathbf{a}_R ds = \frac{q}{\epsilon_0}$$

$$E_R \oint_S ds = E_R (4\pi R^2) = \frac{q}{\epsilon_0}$$

⚡ Important result

$$\mathbf{E} = \mathbf{a}_R E_R = \mathbf{a}_R \frac{q}{4\pi\epsilon_0 R^2} \quad (\text{V/m})$$

- If the charge is not at the origin of the coordinate system



$$\mathbf{E}_P = \mathbf{a}_{qP} \frac{q}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^2}, \quad \mathbf{a}_{qP} = \frac{\mathbf{R} - \mathbf{R}'}{|\mathbf{R} - \mathbf{R}'|}$$

⚡ Important result

$$\mathbf{E}_P = \frac{q(\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3} \quad (\text{V/m})$$

- When a point charge \$q_2\$ is placed in the electric field of another point charge \$q_1\$, the force acting on it is

⚡ Important result | Mathematical statement of Coulomb's law

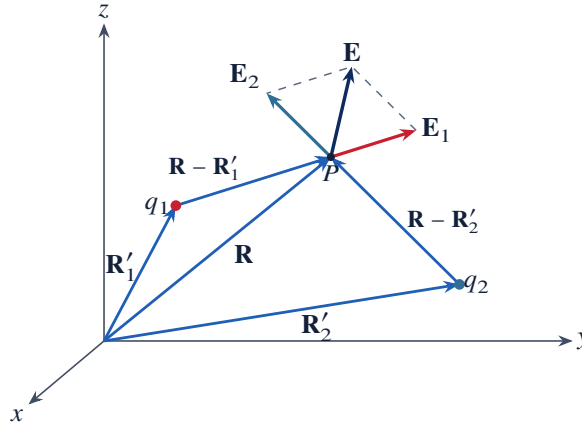
$$\mathbf{F}_{12} = q_2 \mathbf{E}_{12} = \mathbf{a}_R \frac{q_1 q_2}{4\pi\epsilon_0 R^2} \quad (\text{N})$$



► The electric field intensity of a set of discrete charges

⚡ Important result

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k(\mathbf{R} - \mathbf{R}'_k)}{|\mathbf{R} - \mathbf{R}'_k|^3} \quad (\text{V/m})$$



► **The electric field intensity of a continuous charge distribution**

- The field of a differential charge dq is

$$d\mathbf{E} = \frac{dq(\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Line charge distribution**

$$dq = \rho_\ell d\ell' \quad \mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Surface charge distribution**

$$dq = \rho_s ds' \quad \mathbf{E} = \int_{S'} \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Volume charge distribution**

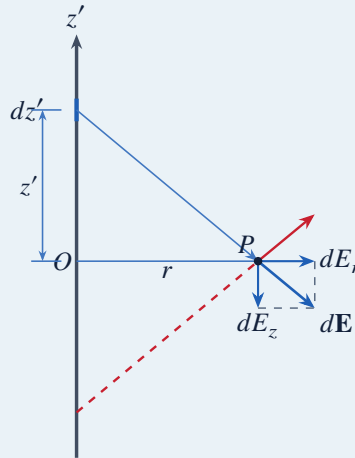
$$dq = \rho_v dv' \quad \mathbf{E} = \int_{V'} \frac{\rho_v dv' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Steps in computing the electric field intensity of a continuous charge distribution**

- Choosing an appropriate coordinate system
- Correctly determining the differential element $d\ell'$, ds' and dv' (note that in these relations the magnitudes of the differential length and surface vectors are used)
- Correctly determining the position vector of the field point ($\mathbf{R}$)
- Correctly determining the position vector of the source point ($\mathbf{R}'$)
- Evaluating the integral
 - Note that we integrate with respect to the primed coordinates.

Example 2-1

Determine the electric field intensity of an infinitely long straight line charge of uniform density ρ_ℓ .



► Solution.

$$\mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

► Choosing cylindrical coordinates

$$d\ell' = dz' \quad \mathbf{R} = r\mathbf{a}_r \quad \mathbf{R}' = z'\mathbf{a}_z$$

$$\mathbf{E} = \frac{\rho_\ell}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{r\mathbf{a}_r - z'\mathbf{a}_z}{(r^2 + z'^2)^{3/2}} dz'$$

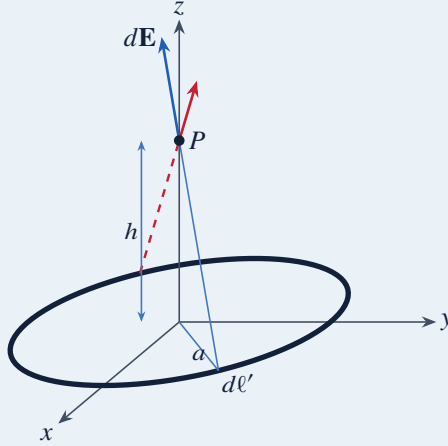
► It is intuitively clear that the field has no component along z .

$$\mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{\rho_\ell r}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dz'}{(r^2 + z'^2)^{3/2}}$$

$$\mathbf{E} = \mathbf{a}_r \frac{\rho_\ell}{2\pi\epsilon_0 r} \quad (\text{V/m})$$

Example 2-2

A ring of radius a carries a line charge density ρ_ℓ . Find the electric field intensity on the axis of the ring at a distance h from it.



► Solution.

$$\mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

► Choosing cylindrical coordinates

$$d\ell' = a d\phi' \quad \mathbf{R} = h\mathbf{a}_z \quad \mathbf{R}' = a\mathbf{a}_{r'}$$

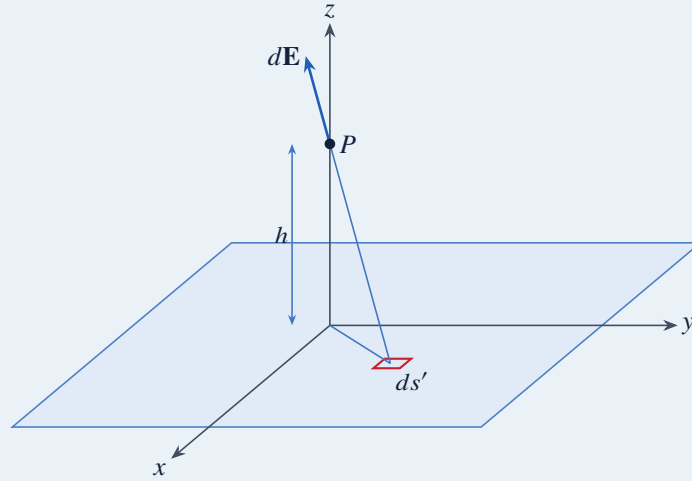
$$\mathbf{E} = \frac{\rho_\ell}{4\pi\epsilon_0} \int_0^{2\pi} \frac{h\mathbf{a}_z - a\mathbf{a}_{r'}}{(a^2 + h^2)^{3/2}} a d\phi'$$

► It is intuitively clear that the field has no component along r .

$$\mathbf{E} = \frac{\rho_\ell ah}{4\pi\epsilon_0 (a^2 + h^2)^{3/2}} \mathbf{a}_z \int_0^{2\pi} d\phi' = \frac{\rho_\ell ah}{2\epsilon_0 (a^2 + h^2)^{3/2}} \mathbf{a}_z$$

Example 2-3

Determine the electric field intensity of an infinite planar charge with a uniform surface charge density ρ_s .



► **Solution.**

$$\mathbf{E} = \int_{S'} \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

► Choosing cylindrical coordinates

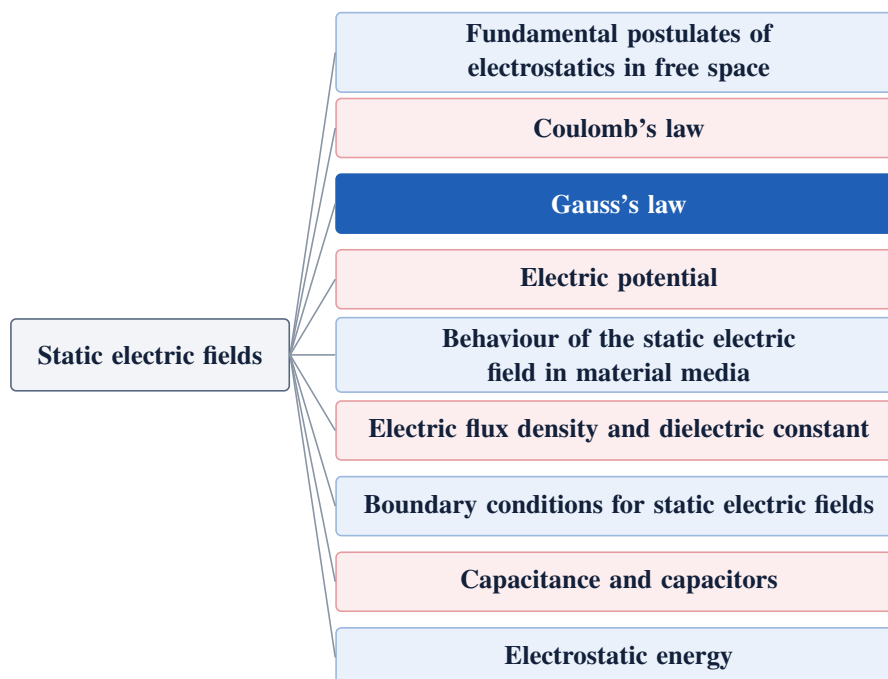
$$ds' = r' dr' d\phi' \quad \mathbf{R} = h\mathbf{a}_z \quad \mathbf{R}' = r'\mathbf{a}_{r'}$$

$$\mathbf{E} = \frac{\rho_s}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h\mathbf{a}_z - r'\mathbf{a}_{r'}}{(h^2 + r'^2)^{3/2}} r' dr' d\phi'$$

► It is intuitively clear that the field has no component along r .

$$\mathbf{E} = \frac{\rho_s h \mathbf{a}_z}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{r' dr' d\phi'}{(h^2 + r'^2)^{3/2}} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z$$

2.5 Gauss's Law



⚡ Important result

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

- Q : the total charge contained in the volume V enclosed by the surface S
- The total outward flux of the electric field intensity through any closed surface equals the total charge enclosed in the surface divided by ϵ_0 .
- The surface S can be any closed surface, chosen for convenience.

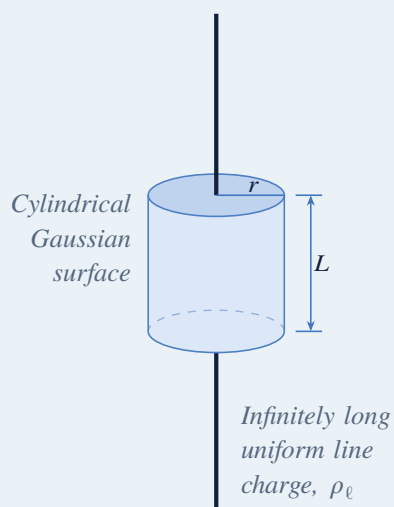
📌 Remark

Gauss's law always holds, but in using it to determine the electric field intensity the following point must be observed.

- The basis of applying Gauss's law to determine the electric field intensity lies, first, in recognizing the conditions of symmetry and, second, in choosing an appropriate surface over which the normal component of $\mathbf{E}$ due to a given charge distribution is constant (the Gaussian surface).

Example 2-4

Using Gauss's law, determine the electric field intensity of an infinitely long straight line charge of uniform density ρ_ℓ at a distance r from it.



► Solution.

$$\mathbf{E} = \mathbf{a}_r E_r$$

► Choosing a cylinder of length L and radius r as the Gaussian surface

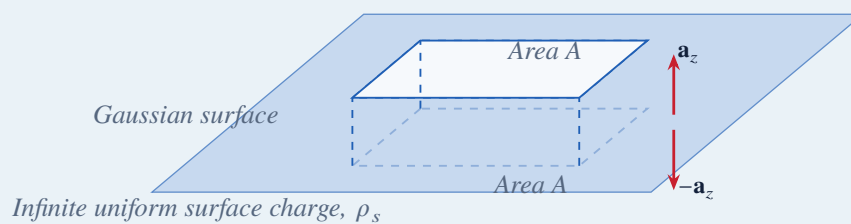
- For the top and bottom faces: $\mathbf{E} \cdot d\mathbf{s} = 0$
- For the side surface: $d\mathbf{s} = \mathbf{a}_r r d\phi dz$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \int_0^L \int_0^{2\pi} E_r r d\phi dz = 2\pi r L E_r$$

$$Q = \rho_\ell L \quad 2\pi r L E_r = \frac{\rho_\ell L}{\epsilon_0} \quad \mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{\rho_\ell}{2\pi \epsilon_0 r}$$

Example 2-5

Determine the electric field intensity of an infinite planar charge with a uniform surface charge density ρ_s .



► Solution.

$$\mathbf{E} = \pm E_z \mathbf{a}_z$$

- Choosing a box as the Gaussian surface
- For the top face

$$\mathbf{E} \cdot d\mathbf{s} = (\mathbf{a}_z E_z) \cdot (\mathbf{a}_z ds) = E_z ds$$

- For the bottom face

$$\mathbf{E} \cdot d\mathbf{s} = (-\mathbf{a}_z E_z) \cdot (-\mathbf{a}_z ds) = E_z ds$$

- For the side faces

$$\mathbf{E} \cdot d\mathbf{s} = 0$$

- Therefore

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = 2E_z \int_A ds = 2E_z A$$

$$Q = \rho_s A \quad 2E_z A = \frac{\rho_s A}{\epsilon_0}$$

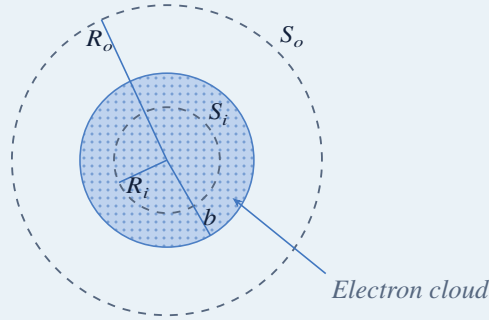
⚡ Important result

$$\mathbf{E} = \mathbf{a}_z E_z = \mathbf{a}_z \frac{\rho_s}{2\epsilon_0}, \quad z > 0$$

$$\mathbf{E} = -\mathbf{a}_z E_z = -\mathbf{a}_z \frac{\rho_s}{2\epsilon_0}, \quad z < 0$$

Example 2-6

Determine the field $\mathbf{E}$ of an electron cloud with volume charge density $\rho = -\rho_0$ in the region $0 \leq R \leq b$ and $\rho = 0$ in the region $R > b$.



► Solution.

$$\mathbf{E} = \mathbf{a}_R E_R$$

► (a) $0 \leq R \leq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

$$\oint_{S_i} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_i} ds = E_R 4\pi R^2$$

$$Q = \int_V \rho dv = -\rho_0 \int_V dv = -\rho_0 \frac{4\pi}{3} R^3 \quad \mathbf{E} = -\mathbf{a}_R \frac{\rho_0}{3\epsilon_0} R$$

► (b) $R \geq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

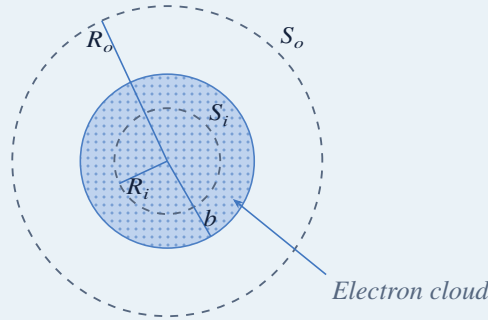
$$\oint_{S_o} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_o} ds = E_R 4\pi R^2$$

$$Q = -\rho_0 \frac{4\pi}{3} b^3 \quad \mathbf{E} = -\mathbf{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}$$

Example 2-7

A spherical charge distribution of radius b has the following density. Find the electric field intensity at all points of space.

$$\rho_v = \begin{cases} \frac{\rho_0 R}{b} & 0 \leq R \leq b \\ 0 & R > b \end{cases}$$



► Solution.

$$\mathbf{E} = \mathbf{a}_R E_R$$

- (a) $0 \leq R \leq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

$$\oint_{S_i} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_i} ds = E_R 4\pi R^2$$

$$Q = \int_v \rho_v dv = \int_0^{2\pi} \int_0^\pi \int_0^R \frac{\rho_0 R}{b} R^2 \sin \theta dR d\theta d\phi = \frac{\rho_0 \pi R^4}{b}$$

$$\mathbf{E} = E_R \mathbf{a}_R = \frac{\rho_0 R^2}{4\epsilon_0 b} \mathbf{a}_R$$

- (b) $R \geq b$

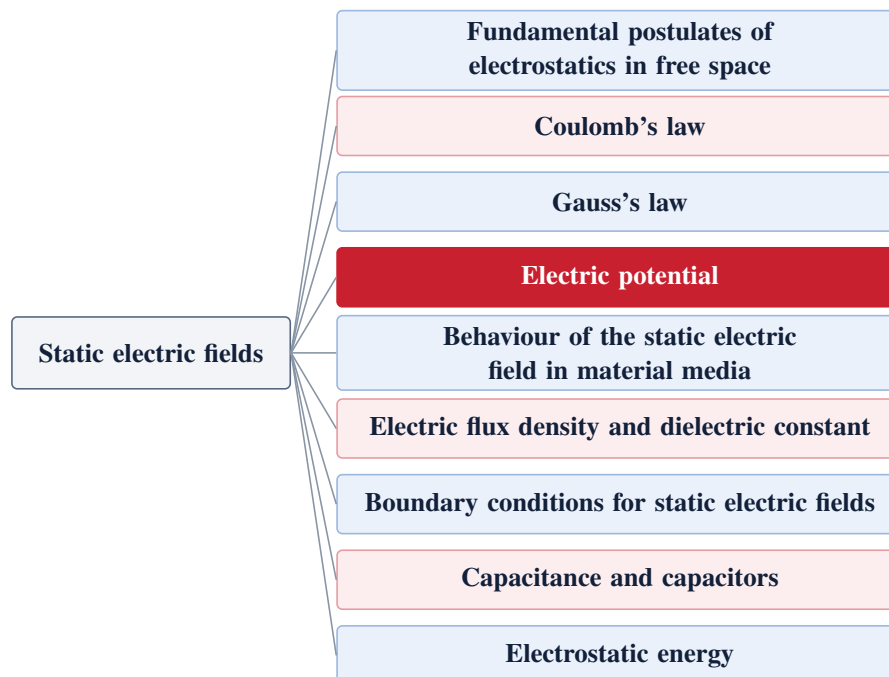
– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

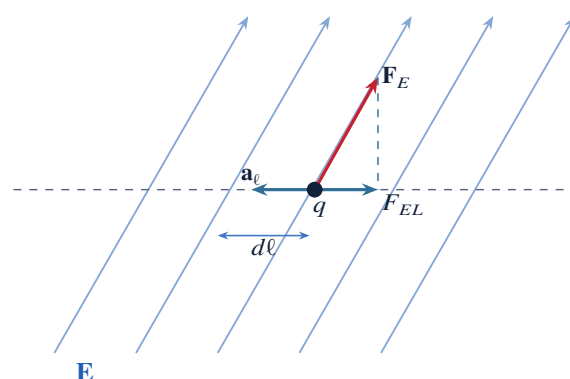
$$\oint_{S_o} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_o} ds = E_R 4\pi R^2 \quad Q = \rho_0 \pi b^3$$

$$\mathbf{E} = E_R \mathbf{a}_R = \frac{\rho_0 b^3}{4\epsilon_0 R^2} \mathbf{a}_R$$

2.6 Electric Potential



- What is the concept of the electric potential difference between two points and of the electric potential of a point?
- How is the electric potential of a point charge, of a number of discrete point charges and of a continuous charge distribution computed?
- How can the electric field intensity be computed from the electric potential, and what is the advantage of this over computing the electric field intensity directly?
- Suppose we want to move a point charge q , located in the electric field $\mathbf{E}$, by $d\ell$.



- The force exerted on the charge by the electric field:

$$\mathbf{F}_E = q\mathbf{E}$$

- The magnitude of the component of this force along $d\ell$:

$$F_{EL} = \mathbf{F}_E \cdot \mathbf{a}_\ell = q\mathbf{E} \cdot \mathbf{a}_\ell$$

- The force that must be applied to the charge:

$$F_{\text{apply}} = -F_{EL} = -q\mathbf{E} \cdot \mathbf{a}_\ell$$

- ▶ The work done in moving the charge by $d\ell$:

$$dW = F_{\text{apply}} d\ell = -q\mathbf{E} \cdot d\ell$$

- ▶ The work done in moving the charge from point P_1 to point P_2 :

$$W = -q \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

$$\nabla \times \mathbf{E} = 0 \Rightarrow \mathbf{E} = -\nabla V$$

$$\nabla V \cdot \mathbf{a}_\ell = \frac{dV}{d\ell}$$

$$q = +1 \text{ C} \quad W = \int_{P_1}^{P_2} \nabla V \cdot d\ell \quad \Rightarrow \quad W = \int_{P_1}^{P_2} dV = V_2 - V_1$$

Definition | Electric potential difference

The electric potential difference between two points is the work done by an external source in moving a unit positive charge from one point to the other.

$$V_2 - V_1 = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

- ▶ It does not depend on the path of integration
- ▶ In many cases a reference point at infinity with zero potential is chosen, and the potential differences of the various points are expressed with respect to this reference point.

Important result

$$V_P = - \int_{\infty}^P \mathbf{E} \cdot d\ell$$

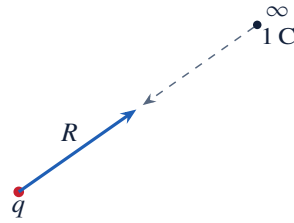
Remark | Remark 1

The minus sign in the potential relation is needed for consistency with the fact that moving against the direction of the field increases the electric potential.

Remark | Remark 2

The gradient of V is normal to the surfaces of constant V ; consequently the electric field lines are normal to the equipotential surfaces.

- ▶ The electric potential of a point charge q at a distance R from it



$$V = - \int_{\infty}^R \left(\frac{q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \right) \cdot dR \mathbf{a}_R = \frac{q}{4\pi\epsilon_0 R}$$

- ▶ The electric potential of n discrete point charges

$$V = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k}{|\mathbf{R} - \mathbf{R}'_k|}$$

- ▶ $\mathbf{R}$: position vector of the field point; $\mathbf{R}'_k$: position vectors of the source points

- ▶ The electric potential of a continuous charge distribution

- Line charge distribution

$$V = \frac{1}{4\pi\epsilon_0} \int_{L'} \frac{\rho_\ell d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

- Surface charge distribution

$$V = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_s ds'}{|\mathbf{R} - \mathbf{R}'|}$$

- Volume charge distribution

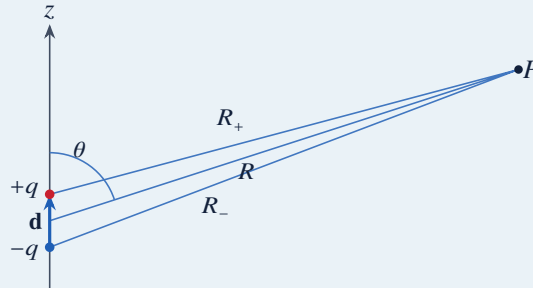
$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_v dv'}{|\mathbf{R} - \mathbf{R}'|}$$

Example 2-8

Find the electric potential and the electric field intensity of an electric dipole at an arbitrary point very far from it.

► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^2 \frac{q_k}{|\mathbf{R} - \mathbf{R}'_k|} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\mathbf{R} - \mathbf{d}/2|} + \frac{-q}{|\mathbf{R} + \mathbf{d}/2|} \right]$$



$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_+} - \frac{1}{R_-} \right)$$

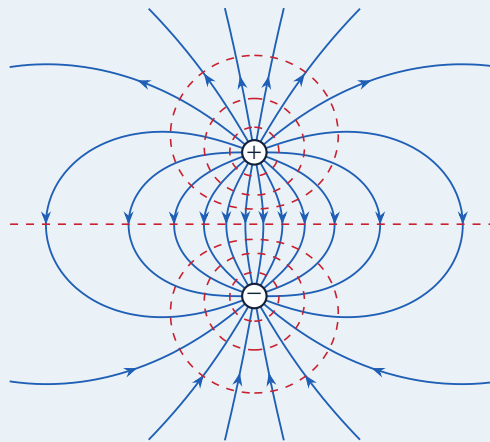
$$\frac{1}{R_+} \cong \left(R - \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 + \frac{d}{2R} \cos \theta \right)$$

$$\frac{1}{R_-} \cong \left(R + \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 - \frac{d}{2R} \cos \theta \right)$$

► $\mathbf{p} = q\mathbf{d}$: the electric dipole moment vector

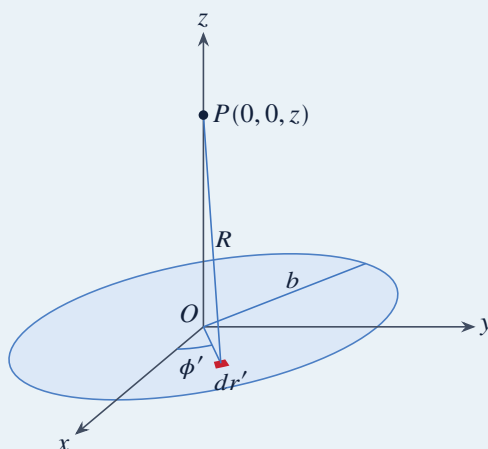
$$V = \frac{qd \cos \theta}{4\pi\epsilon_0 R^2} \quad \mathbf{d} = d\mathbf{a}_z \quad \Rightarrow \quad V = \frac{q\mathbf{d} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} \quad \mathbf{p} = q\mathbf{d} \quad \Rightarrow \quad V = \frac{\mathbf{p} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2}$$

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\mathbf{a}_R \frac{\partial V}{\partial R} - \mathbf{a}_\theta \frac{1}{R} \frac{\partial V}{\partial \theta} \\ &= \frac{p}{4\pi\epsilon_0 R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta) \end{aligned}$$



Example 2-9

Find the electric potential and the electric field intensity of a charged disk of radius b with surface charge density ρ_s on its axis.



► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_s ds'}{|\mathbf{R} - \mathbf{R}'|}$$

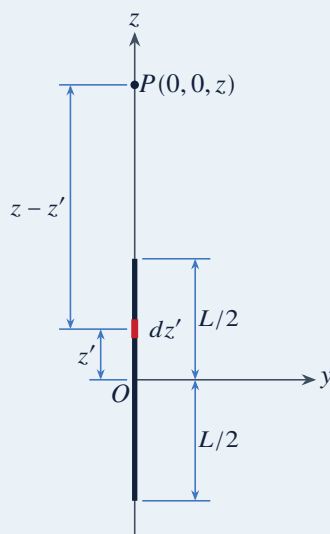
$$ds' = r' dr' d\phi' \quad \mathbf{R} = z\mathbf{a}_z \quad \mathbf{R}' = r'\mathbf{a}_{r'}$$

$$V = \frac{\rho_s}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^b \frac{r' dr' d\phi'}{\sqrt{z^2 + r'^2}} = \frac{\rho_s}{2\epsilon_0} \left[\sqrt{z^2 + b^2} - z \right] \quad z > 0$$

$$\mathbf{E} = -\nabla V = \mathbf{a}_z \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + b^2}} \right] \quad z > 0$$

Example 2-10

Find the electric potential and the electric field intensity of a line charge of length L and charge density ρ_ℓ along its extension.



► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \int_{L'} \frac{\rho_\ell d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

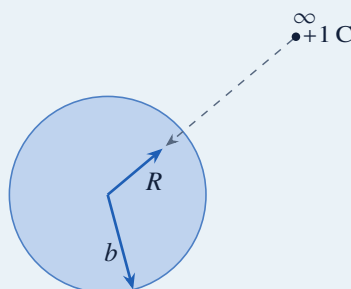
$$d\ell' = dz' \quad \mathbf{R} = z\mathbf{a}_z \quad \mathbf{R}' = z'\mathbf{a}_z$$

$$V = \frac{\rho_\ell}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{dz'}{z - z'} = \frac{\rho_\ell}{4\pi\epsilon_0} \ln \left[\frac{z + L/2}{z - L/2} \right] \quad z > L/2$$

$$\mathbf{E} = -\nabla V = -\mathbf{a}_z \frac{dV}{dz} = \mathbf{a}_z \frac{\rho_\ell L}{4\pi\epsilon_0 [z^2 - (L/2)^2]} \quad z > L/2$$

Example 2-11

A volume charge of density $\rho = -\rho_0$ exists in the region $0 \leq R \leq b$. What is the electric potential at an arbitrary point inside this volume?



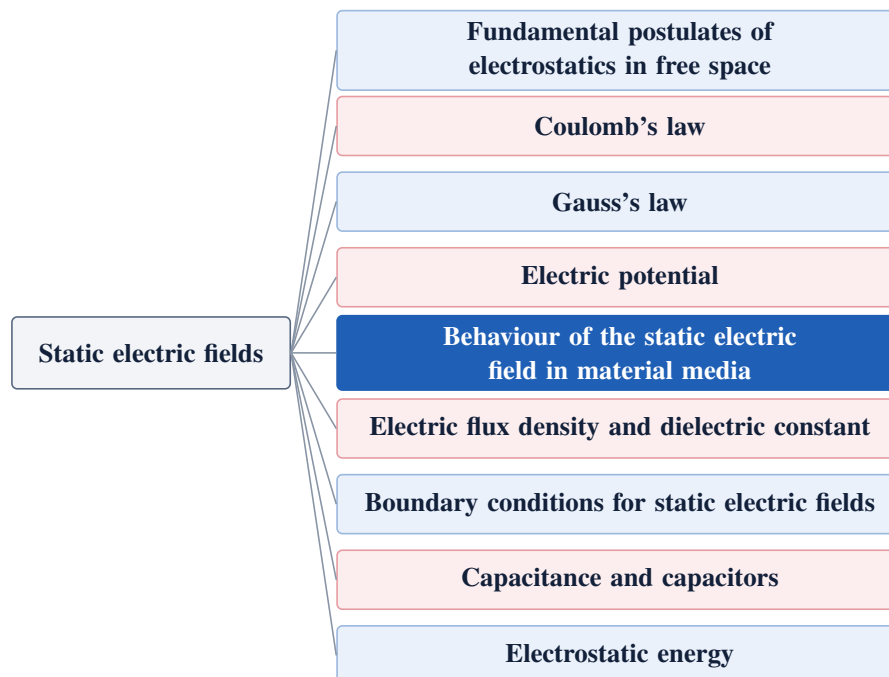
► **Solution.**

$$\mathbf{E} = -\mathbf{a}_R \frac{\rho_0}{3\epsilon_0} R, \quad 0 \leq R \leq b$$

$$\mathbf{E} = -\mathbf{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}, \quad R \geq b$$

$$\begin{aligned} V &= -\int_{\infty}^R \mathbf{E} \cdot d\ell = -\int_{\infty}^b \frac{-\rho_0 b^3}{3\epsilon_0 R^2} \mathbf{a}_R \cdot dR \mathbf{a}_R - \int_b^R \frac{-\rho_0 R}{3\epsilon_0} \mathbf{a}_R \cdot dR \mathbf{a}_R \\ &= -\frac{\rho_0 b^2}{3\epsilon_0} - \frac{\rho_0}{6\epsilon_0} (b^2 - R^2) \end{aligned}$$

2.7 Behaviour of the Static Electric Field in Material Media

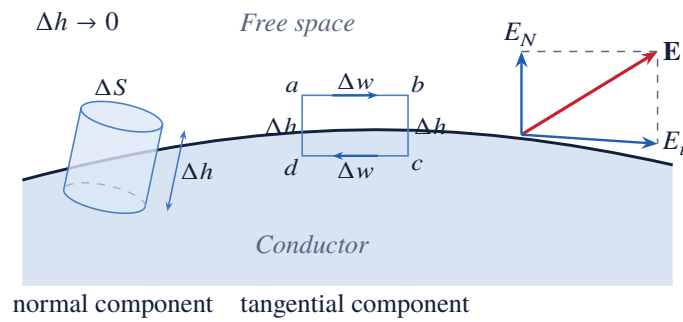


- Classification of materials according to their electrical properties
 - Conductors
 - Semiconductors
 - Insulators (dielectrics)
- In conductors the electrons of the outer shells of the atoms are held by a weak force and pass easily from one atom to another.
- But the electrons in the atoms of insulators are tightly bound to their orbits and under normal conditions cannot be detached.
- The electrical properties of semiconductors lie between those of conductors and insulators.
- How does the static electric field intensity behave in the presence of a conductor?
- How does the static electric field intensity behave in the presence of a dielectric?
- What is the concept of electric flux density, and how is it related to the electric field intensity?
- What is the concept of the dielectric constant?
- How does the static electric field intensity behave at the boundary between two different media?

2.8 Conductors in Static Electric Field

- If we release charges inside a conductor, a field forms in it that drives the charges apart and to the surface.
 - Under static conditions the charge and the electric field inside a conductor are zero.

$$\rho = 0 \quad \mathbf{E} = 0$$
 - The time needed for the charges to distribute themselves on the surface of the conductor and reach equilibrium depends on the conductivity of the conductor.
- If under static conditions the electric field on the surface of a conductor had a tangential component, it would produce a tangential force, set the charges in motion, and there would be no charge equilibrium.
 - Therefore, under static conditions the electric field on the surface of a conductor is everywhere normal to the surface. In other words, under static conditions the surface of a conductor is an equipotential surface. In fact, since $\mathbf{E} = 0$ at all points inside the conductor, the whole conductor has the same electric potential.
- **Boundary conditions at a conductor / free-space interface**



- Normal component:

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_n \Delta S = \frac{\rho_s \Delta S}{\epsilon_0} \quad \Rightarrow \quad E_n = \frac{\rho_s}{\epsilon_0}$$

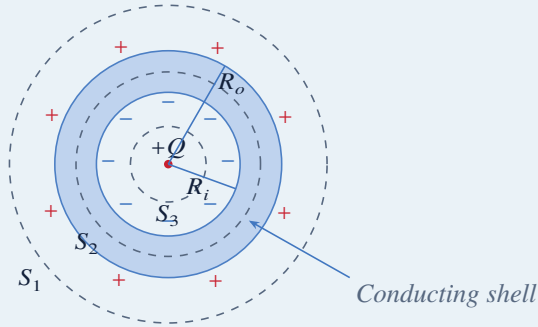
- Tangential component:

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$E_t \Delta w - E_{N, \text{at } b} \frac{1}{2} \Delta h + E_{N, \text{at } a} \frac{1}{2} \Delta h = 0 \quad \Rightarrow \quad E_t = 0$$

Example 2-12

A point charge $+Q$ is at the centre of a conducting spherical shell of inner radius R_i and outer radius R_o . Find $\mathbf{E}$ and V in all space.



► Solution.

$$\mathbf{E} = E_R \mathbf{a}_R$$

► For $R > R_o$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_{R1} 4\pi R^2 = \frac{Q}{\epsilon_0}$$

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2} \quad V_1 = -\int_{\infty}^R (E_{R1}) dR = \frac{Q}{4\pi\epsilon_0 R}$$

► For $R_i < R < R_o$

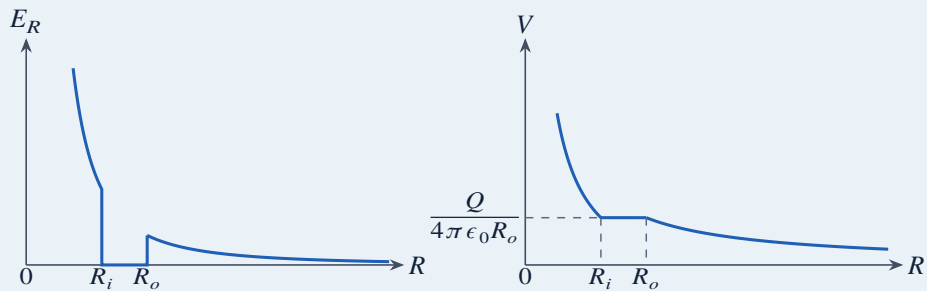
$$E_{R2} = 0 \quad V_2 = V_1|_{R=R_o} = \frac{Q}{4\pi\epsilon_0 R_o}$$

► For $R < R_i$

$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}$$

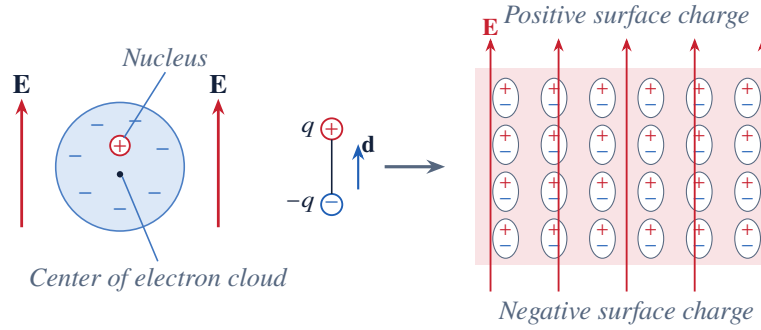
$$V_3 = -\int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^{R_i} E_{R2} dR - \int_{R_i}^R E_{R3} dR$$

$$V_3 = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R} + \frac{1}{R_o} - \frac{1}{R_i} \right)$$



2.9 Dielectrics in Static Electric Field

- ▶ Ideal dielectrics have no free charges. As a result, their charge density and internal electric field are not zero, as they are in conductors.
- ▶ Dielectrics have **bound charges**; applying an external field to a dielectric creates electric dipoles and polarizes the dielectric material.
- ▶ The induced electric dipoles change the electric field inside and outside the dielectric material.



▶ Equivalent charge distributions in polarized dielectrics

- To analyse the macroscopic effect of the induced dipoles, we define the polarization vector $\mathbf{P}$ as follows

Definition

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{p}_k}{\Delta v} \quad (\text{C/m}^2)$$

- ▶ n : number of dipoles per unit volume; $\mathbf{p}_k$: dipole moment
- ▶ If $d\mathbf{p}$ is the dipole moment of a small volume dv' , then

$$d\mathbf{p} = \mathbf{P} dv' \quad dV = \frac{\mathbf{P} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} dv' \quad V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\mathbf{P} \cdot \mathbf{a}_R}{R^2} dv'$$

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\mathbf{P} \cdot \mathbf{a}'_n}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{(-\nabla' \cdot \mathbf{P})}{R} dv'$$

- ▶ The electric potential, and hence the electric field intensity, of a polarized dielectric can be computed from the effect of two charge distributions, a surface one and a volume one, with the following densities, called the **polarization charge densities** or **bound charge densities**

Important result

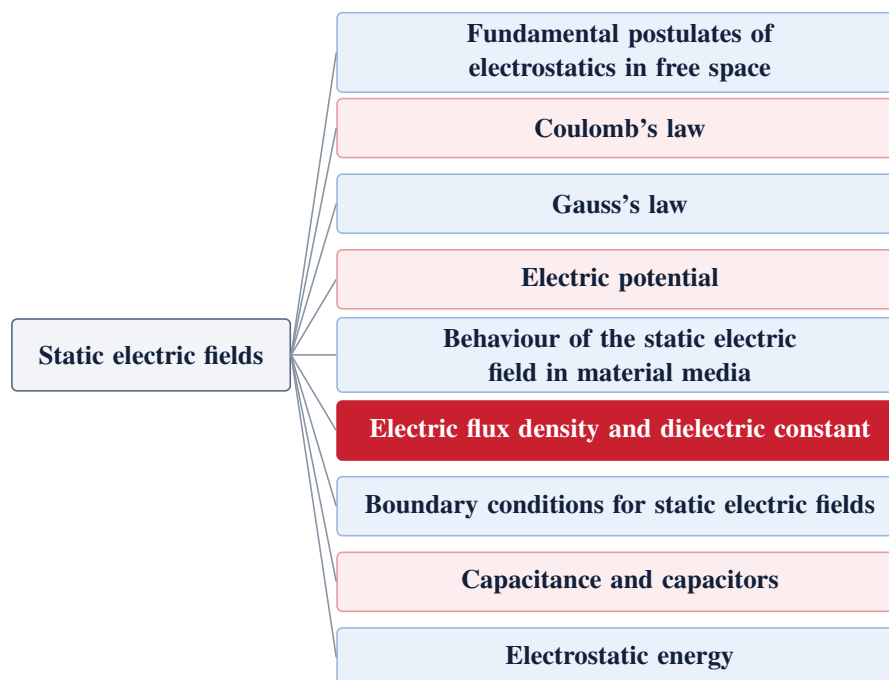
$$\begin{aligned} \rho_{ps} &= \mathbf{P} \cdot \mathbf{a}_n & \rho_p &= -\nabla \cdot \mathbf{P} \\ V &= \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\rho_{ps}}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_p}{R} dv' \end{aligned}$$

① Remark

Since we are dealing with an electrically neutral dielectric body, the total charge of the body after polarization must still be zero

$$\begin{aligned}\text{Total charge} &= \oint_S \rho_{ps} ds + \int_V \rho_p dv \\ &= \oint_S \mathbf{P} \cdot \mathbf{a}_n ds - \int_V \nabla \cdot \mathbf{P} dv = 0\end{aligned}$$

2.10 Electric Flux Density



- Since a dielectric gives rise to an equivalent volume charge density ρ_p , we expect the electric field intensity due to a given charge distribution in a dielectric to differ from the electric field intensity in free space.

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho + \rho_p)$$

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho$$

- The electric flux density or electric displacement $\mathbf{D}$ is defined as follows

≡ Definition

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (\text{C/m}^2)$$

- Using the vector $\mathbf{D}$, the relation between the electric field and the free charge distributions in any medium can be written without having to deal with the polarization vector $\mathbf{P}$ or the polarization charge densities.

⚡ Important result

$$\nabla \cdot \mathbf{D} = \rho \quad (\text{C/m}^3)$$

- Integral form

$$\int_V \nabla \cdot \mathbf{D} dv = \int_V \rho dv$$

⚡ Important result | Another form of Gauss's law

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad (\text{C})$$

- The fundamental equations of electrostatics in any medium

⚡ Important result

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = 0 \end{cases}$$

2.11 Dielectric Constant

- The relation between the polarization vector and the electric field intensity

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

- χ_e : electric susceptibility
- The relation between the electric field intensity and the electric flux density

⚡ Important result

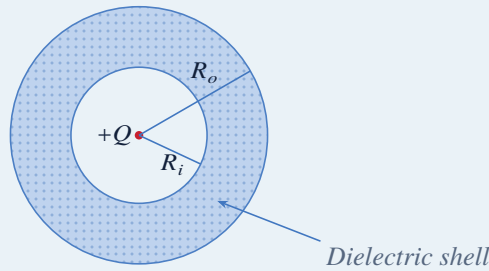
$$\begin{aligned} \mathbf{D} &= \epsilon_0(1 + \chi_e)\mathbf{E} \\ &= \epsilon_0\epsilon_r\mathbf{E} = \epsilon\mathbf{E} \quad (\text{C/m}^2) \end{aligned}$$

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$$

- ▶ ϵ_r : a dimensionless constant called the relative permittivity or dielectric constant
- ▶ A dielectric medium is **linear** if ϵ_r is independent of $\mathbf{E}$, **homogeneous** if ϵ_r is independent of the space coordinates, and isotropic if ϵ_r is a scalar.
 - A linear, homogeneous and isotropic medium is called a simple medium.
 - In anisotropic media the dielectric constant is a matrix. In such media $\mathbf{D}$ and $\mathbf{E}$ are not necessarily in the same direction.

Example 2-13

A point charge $+Q$ is at the centre of a dielectric spherical shell of inner radius R_i and outer radius R_o . The dielectric constant of the shell is ϵ_r . Find $\mathbf{E}$, V , $\mathbf{D}$ and $\mathbf{P}$ in all space.



Solution.

$$\mathbf{E} = E_R \mathbf{a}_R$$

- ▶ For $R > R_o$

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2} \quad V_1 = \frac{Q}{4\pi\epsilon_0 R}$$

$$D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2} \quad P_{R1} = 0$$

- ▶ For $R_i < R < R_o$

$$D_{R2} = \frac{Q}{4\pi R^2}$$

$$E_{R2} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} = \frac{Q}{4\pi\epsilon R^2}$$

$$P_{R2} = \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R^2}$$

$$\begin{aligned} V_2 &= - \int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^R E_{R2} dR \\ &= V_1 \Big|_{R=R_o} - \frac{Q}{4\pi\epsilon} \int_{R_o}^R \frac{1}{R^2} dR \\ &= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} + \frac{1}{\epsilon_r R} \right] \end{aligned}$$

- ▶ For $R < R_i$

$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2} \quad D_{R3} = \frac{Q}{4\pi R^2} \quad P_{R3} = 0$$

$$\begin{aligned} V_3 &= V_2|_{R=R_i} - \int_{R_i}^R E_{R3} dR \\ &= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_i} + \frac{1}{R} \right] \end{aligned}$$

$$\begin{aligned} \rho_{ps}|_{R=R_i} &= \mathbf{P} \cdot (-\mathbf{a}_R)|_{R=R_i} = -P_{R2}|_{R=R_i} \\ &= -\left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_i^2} \end{aligned}$$

$$\begin{aligned} \rho_{ps}|_{R=R_o} &= \mathbf{P} \cdot \mathbf{a}_R|_{R=R_o} = P_{R2}|_{R=R_o} \\ &= \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_o^2} \end{aligned}$$

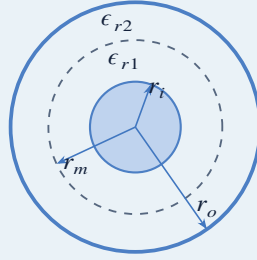
$$\begin{aligned} \rho_p &= -\nabla \cdot \mathbf{P} \\ &= -\frac{1}{R^2} \frac{\partial}{\partial R} (R^2 P_{R2}) = 0 \end{aligned}$$

2.12 Dielectric Strength

- An external electric field causes small displacements of the bound charges and creates electric dipoles.
- If the electric field is strong, it completely detaches electrons from the molecules, accelerates them and makes them collide with the molecular lattice structure.
- In this case the material may become conducting and large currents may arise.
 - This phenomenon is called **dielectric breakdown**.
- The maximum electric field intensity that a dielectric material can withstand without breakdown is called the **dielectric strength** of the material.
 - For example, the dielectric strength of air is 3 kV/mm.

Example 2-14

When a coaxial cable is used to carry electric power, the radius of the inner conductor is determined by the load current, and the overall size by the voltage and the insulating material used. Suppose that the radius of the inner conductor is 0.4 cm and that two layers, rubber with relative permittivity $\epsilon_{rr} = 3.2$ and polystyrene with relative permittivity $\epsilon_{rp} = 2.6$, are used as the insulating material. Design this cable so that it can operate at a voltage of 20 kV. To avoid breakdown due to abnormal external conditions, the maximum electric field intensity in the insulating materials must not exceed 25% of their dielectric strengths. (The dielectric strength of rubber is 25×10^6 V/m and that of polystyrene is 20×10^6 V/m.)



► Solution.

- The aim is to determine the order of the insulating layers and the values of r_o and r_m .
- Assuming that the free charge density on the surface of the inner conductor is ρ_s , Gauss's law gives

$$D_r 2\pi r L = \rho_s 2\pi r_i L \quad D_r = \rho_s \frac{r_i}{r} \quad E_r = \frac{\rho_s}{\epsilon_0 \epsilon_{r1,2}} \frac{r_i}{r}$$

$$\text{Rubber} \quad \max\{E_{rr}\} = 0.25 \times 25 \times 10^6 \text{ V/m} = \frac{\rho_s}{\epsilon_0 \epsilon_{rr}} \frac{r_i}{r_{r \min}} \quad r_{r \min} = r_i \text{ or } r_m$$

$$\text{Polystyrene} \quad \max\{E_{rp}\} = 0.25 \times 20 \times 10^6 \text{ V/m} = \frac{\rho_s}{\epsilon_0 \epsilon_{rp}} \frac{r_i}{r_{p \min}} \quad r_{p \min} = r_i \text{ or } r_m$$

$$\frac{25}{20} = \frac{\epsilon_{rp}}{\epsilon_{rr}} \times \frac{r_{p \min}}{r_{r \min}} \quad r_{p \min} > r_{r \min} \Rightarrow \begin{cases} r_{r \min} = r_i \\ r_{p \min} = r_m \end{cases}$$

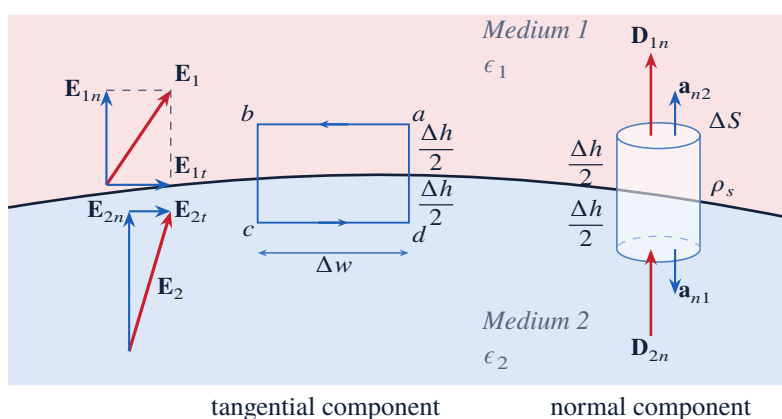
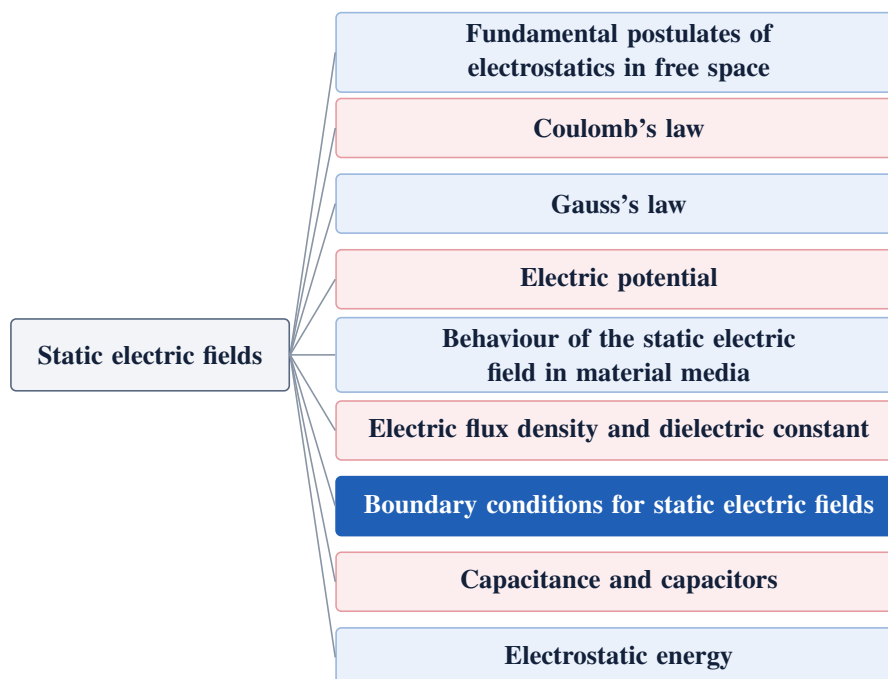
$$\frac{25}{20} = \frac{2.6}{3.2} \times \frac{r_m}{0.4} \Rightarrow r_m = 0.616 \text{ cm}$$

$$\begin{aligned} - \int_{r_o}^{r_m} E_{rp} dr - \int_{r_m}^{r_i} E_{rr} dr &= \frac{\rho_s r_i}{\epsilon_0} \left(\frac{1}{\epsilon_{rp}} \ln \left(\frac{r_o}{r_m} \right) + \frac{1}{\epsilon_{rr}} \ln \left(\frac{r_m}{r_i} \right) \right) \\ &= 20000 \text{ V} \end{aligned}$$

$$\frac{\rho_s r_i}{\epsilon_0} = 0.25 \times 25 \times 10^6 \epsilon_{rr} r_i = 8 \times 10^4$$

$$\Rightarrow r_o = 0.832 \text{ cm}$$

2.13 Boundary Conditions for Static Electric Fields



► Tangential component:

$$\oint_{ab c d a} \mathbf{E} \cdot d\boldsymbol{\ell} = \mathbf{E}_1 \cdot \Delta \mathbf{w} + \mathbf{E}_2 \cdot (-\Delta \mathbf{w}) = E_{1t} \Delta w - E_{2t} \Delta w = 0$$

⚡ Important result

$$E_{1t} = E_{2t} \qquad \frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

- Normal component:

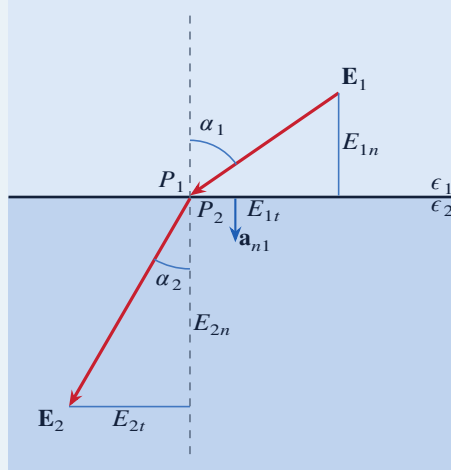
$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{s} &= (\mathbf{D}_1 \cdot \mathbf{a}_{n2} + \mathbf{D}_2 \cdot \mathbf{a}_{n1}) \Delta S \\ &= \mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) \Delta S \\ &= \rho_s \Delta S\end{aligned}$$

⚡ Important result

$$\mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \qquad \mathbf{a}_{n2} \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \rho_s$$

Example 2-15

Two dielectric media with permittivities ϵ_1 and ϵ_2 are separated by a charge-free boundary as shown in the figure below. The electric field intensity in medium 1 at the point P_1 has magnitude E_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at the point P_2 in region 2.



► **Solution.**

$$E_{1t} = E_{2t} \quad \Rightarrow \quad E_2 \sin \alpha_2 = E_1 \sin \alpha_1$$

$$\mathbf{a}_{n2} \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = 0 \quad \Rightarrow \quad \epsilon_2 E_2 \cos \alpha_2 = \epsilon_1 E_1 \cos \alpha_1$$

⚡ Important result

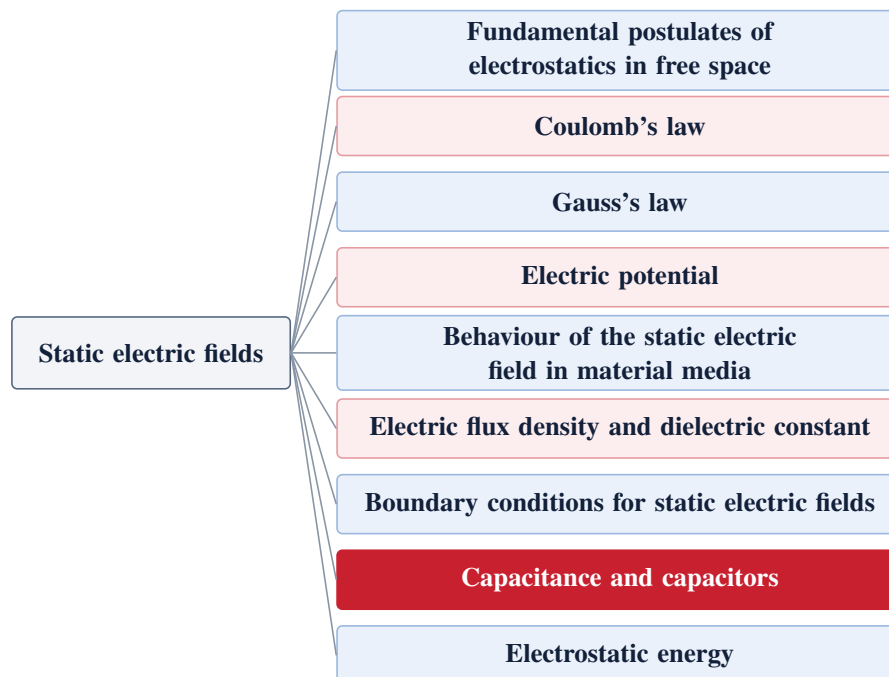
$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\epsilon_2}{\epsilon_1}$$

$$\begin{aligned} E_2 &= \sqrt{E_{2t}^2 + E_{2n}^2} = \sqrt{(E_2 \sin \alpha_2)^2 + (E_2 \cos \alpha_2)^2} \\ &= \left[(E_1 \sin \alpha_1)^2 + \left(\frac{\epsilon_1}{\epsilon_2} E_1 \cos \alpha_1 \right)^2 \right]^{1/2} \end{aligned}$$

⚡ Important result

$$E_2 = E_1 \left[\sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2} \cos \alpha_1 \right)^2 \right]^{1/2}$$

2.14 Capacitance and Capacitors

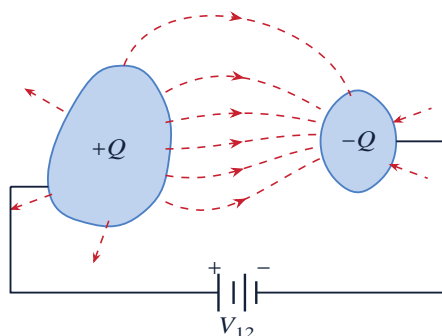


- What kind of element is a capacitor, and what is the concept of capacitance?
- How is the capacitance of various structures computed?
- How can the electric energy stored in a set of charges, or the work needed to assemble a set of charges, be found?
- How can the electric energy stored in capacitive structures be computed?
- Suppose a charge Q is placed on an arbitrary conductor. This charge distributes itself over the surface of the conductor and forms the charge density ρ_s .
- Increasing the total charge by a factor k increases the charge density ρ_s by the factor k , and consequently the potential due to this charge also increases by the factor k .
 - Therefore the ratio of Q to V remains constant

⚡ Important result

$$Q = CV$$

- ▶ The constant of proportionality (C) is called the **capacitance of the isolated conducting body**; it is the electric charge that must be added to the body to raise its electric potential by one unit.
 - Its unit is the coulomb per volt, or farad (F).
- ▶ Two conductors separated by free space or by a dielectric medium are called a capacitor.



- ▶ The capacitance of a capacitor is defined as follows

Definition

$$C = \frac{Q}{V_{12}} \quad (\text{F})$$

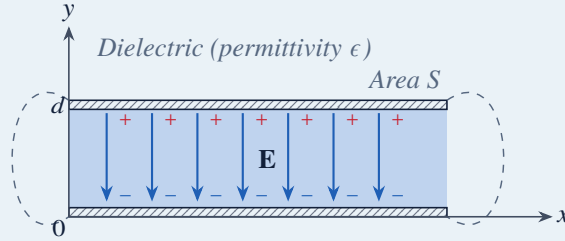
- ▶ The capacitance of a capacitor depends on its geometry and on the permittivity of the medium between the two conductors.
- ▶ Procedure for computing the capacitance of a capacitor
 - Choosing an appropriate coordinate system
 - Placing the charges $+Q$ and $-Q$ on the conductors
 - Finding $\mathbf{E}$ from Q
 - Computing the potential difference between the two conductors from the following relation, from the charge $-Q$ to the charge $+Q$

$$V = - \int_{P-Q}^{P+Q} \mathbf{E} \cdot d\ell$$

- Finding the capacitance from the ratio Q/V

Example 2-16

A parallel-plate capacitor consists of two parallel conducting plates of area S separated by d . The space between the plates is filled with a dielectric of constant permittivity ϵ . Find the capacitance of this capacitor.



► Solution.

- Using Cartesian coordinates
- Placing the charges $+Q$ and $-Q$ on the plates $\rightarrow \rho_s = \frac{Q}{S}$
- Computing the electric field intensity in the space between the two conducting plates with Gauss's law, neglecting the fringing fields

$$\mathbf{E} = -\mathbf{a}_y \frac{\rho_s}{\epsilon} = -\mathbf{a}_y \frac{Q}{\epsilon S}$$

- Computing the potential difference between the conducting plates in terms of Q

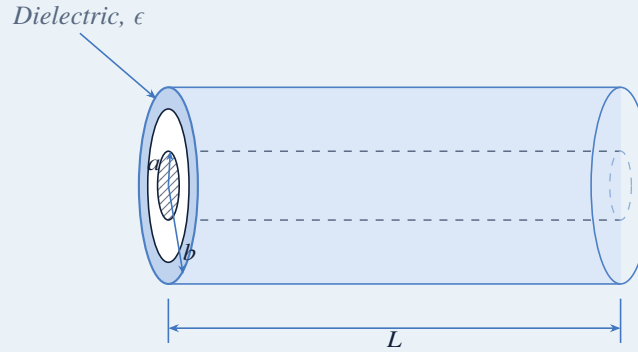
$$V_{12} = - \int_{y=0}^{y=d} \mathbf{E} \cdot d\ell = - \int_0^d \left(-\mathbf{a}_y \frac{Q}{\epsilon S} \right) \cdot (\mathbf{a}_y dy) = \frac{Q}{\epsilon S} d$$

- Determining the capacitance

$$C = \frac{Q}{V_{12}} = \epsilon \frac{S}{d}$$

Example 2-17

A cylindrical capacitor consists of an inner conductor of radius a and an outer conductor of radius b . The space between the two conductors is filled with a dielectric of constant permittivity ϵ , and the length of the capacitor is L . Find the capacitance of this capacitor.



► Solution.

- Using cylindrical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$\mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{Q}{2\pi\epsilon L r}$$

- Computing the potential difference between the conductors in terms of Q

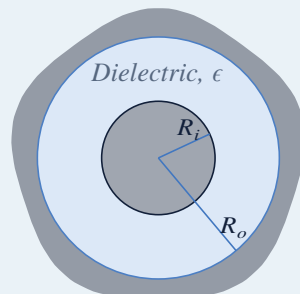
$$\begin{aligned} V_{ab} &= - \int_{r=b}^{r=a} \mathbf{E} \cdot d\ell = - \int_b^a \left(\mathbf{a}_r \frac{Q}{2\pi\epsilon L r} \right) \cdot (\mathbf{a}_r dr) \\ &= \frac{Q}{2\pi\epsilon L} \ln\left(\frac{b}{a}\right) \end{aligned}$$

- Determining the capacitance

$$C = \frac{Q}{V_{ab}} = \frac{2\pi\epsilon L}{\ln\left(\frac{b}{a}\right)}$$

Example 2-18

A spherical capacitor consists of an inner conducting sphere of radius R_i and an outer conducting sphere of radius R_o . The space between the two conductors is filled with a dielectric of constant permittivity ϵ . Determine the capacitance of this capacitor.



► Solution.

- Using spherical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$\mathbf{E} = \mathbf{a}_R E_R = \mathbf{a}_R \frac{Q}{4\pi\epsilon R^2}$$

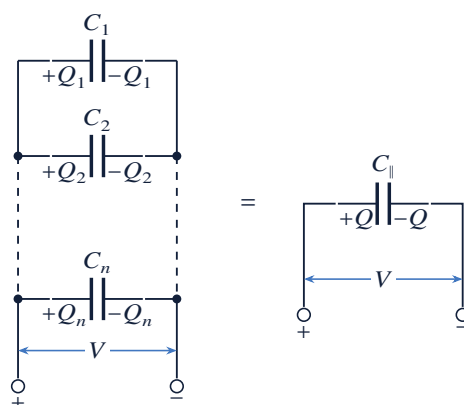
- Computing the potential difference between the conductors in terms of Q

$$V = - \int_{R_o}^{R_i} \mathbf{E} \cdot (\mathbf{a}_R dR) = - \int_{R_o}^{R_i} \frac{Q}{4\pi\epsilon R^2} dR = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_i} - \frac{1}{R_o} \right)$$

- Determining the capacitance

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{\frac{1}{R_i} - \frac{1}{R_o}}$$

► Capacitors connected in parallel



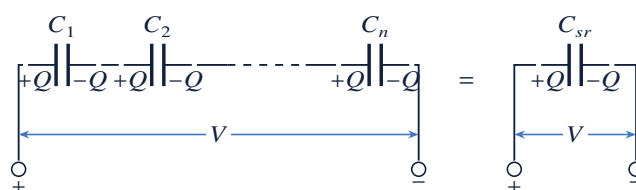
$$Q = Q_1 + Q_2 + \dots + Q_n$$

$$= C_1 V + C_2 V + \dots + C_n V = C_{\parallel} V$$

⚡ Important result

$$C_{\parallel} = C_1 + C_2 + \dots + C_n$$

► Capacitors connected in series



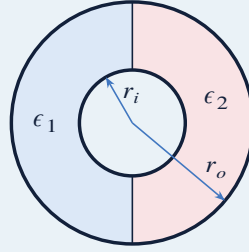
$$V = \frac{Q}{C_{sr}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \dots + \frac{Q}{C_n}$$

⚡ Important result

$$\frac{1}{C_{sr}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

🔗 Example 2-19

A cylindrical capacitor of length L is formed by coaxial cylindrical conducting surfaces of radii r_i and r_o . Two dielectric materials of permittivities ϵ_1 and ϵ_2 fill the space between the conductors as shown in the figure below. Determine the capacitance of this capacitor.



► **Solution.**

► **First method**

- The capacitance of a cylindrical capacitor $\rightarrow C = \frac{2\pi\epsilon L}{\ln\left(\frac{r_o}{r_i}\right)}$
- Since each dielectric fills half of the space, the capacitance of each half is

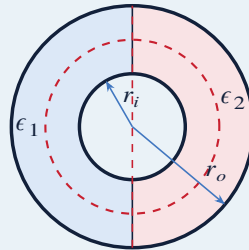
$$C_1 = \frac{\pi\epsilon_1 L}{\ln\left(\frac{r_o}{r_i}\right)} \quad C_2 = \frac{\pi\epsilon_2 L}{\ln\left(\frac{r_o}{r_i}\right)}$$

- Since the two capacitors C_1 and C_2 are in parallel, the total capacitance is

$$C = C_1 + C_2 = \frac{2\pi L}{\ln\left(\frac{r_o}{r_i}\right)} \left(\frac{\epsilon_1 + \epsilon_2}{2} \right)$$

► **Second method**

- Using cylindrical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law



$$\left. \begin{aligned} E_{r1} &= \frac{Q_1}{\pi r \epsilon_1 L} \\ E_{r2} &= \frac{Q_2}{\pi r \epsilon_2 L} \end{aligned} \right\} E_{r1} = E_{r2} \Rightarrow \left. \begin{aligned} \frac{Q_1}{\epsilon_1} &= \frac{Q_2}{\epsilon_2} \\ Q_1 + Q_2 &= Q \end{aligned} \right\} \left\{ \begin{aligned} Q_1 &= \frac{Q\epsilon_1}{\epsilon_1 + \epsilon_2} \\ Q_2 &= \frac{Q\epsilon_2}{\epsilon_1 + \epsilon_2} \end{aligned} \right.$$

- Computing the potential difference between the conductors in terms of Q

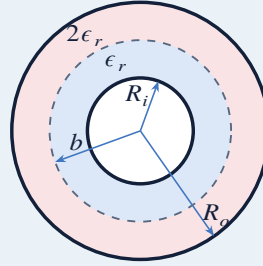
$$V = - \int_{r_o}^{r_i} \frac{Q_1}{\pi r \epsilon_1 L} dr = - \int_{r_o}^{r_i} \frac{Q}{\pi r L (\epsilon_1 + \epsilon_2)} dr = \frac{Q}{\pi L (\epsilon_1 + \epsilon_2)} \ln\left(\frac{r_o}{r_i}\right)$$

- Determining the capacitance

$$C = \frac{2\pi L}{\ln\left(\frac{r_o}{r_i}\right)} \left(\frac{\epsilon_1 + \epsilon_2}{2} \right)$$

Example 2-20

A capacitor is formed by two concentric conducting spherical shells of radii R_i and R_o . The space between the shells is filled with a dielectric of relative permittivity ϵ_r from R_i to b and a dielectric of relative permittivity $2\epsilon_r$ from b to R_o . Determine the capacitance of this capacitor.



► Solution.

► First method

- The capacitance of a spherical capacitor $\rightarrow C = \frac{Q}{V} = \frac{4\pi\epsilon}{\frac{1}{R_i} - \frac{1}{R_o}}$
- This capacitor can be regarded as two capacitors in series, as follows

$$C_1 = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{b}} \quad C_2 = \frac{4\pi\epsilon_0(2\epsilon_r)}{\frac{1}{b} - \frac{1}{R_o}}$$

- Therefore the total capacitance is

$$C = \frac{1}{1/C_1 + 1/C_2} = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o}}$$

► Second method

- Using spherical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$E_{R1} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} \quad E_{R2} = \frac{Q}{4\pi\epsilon_0(2\epsilon_r)R^2}$$

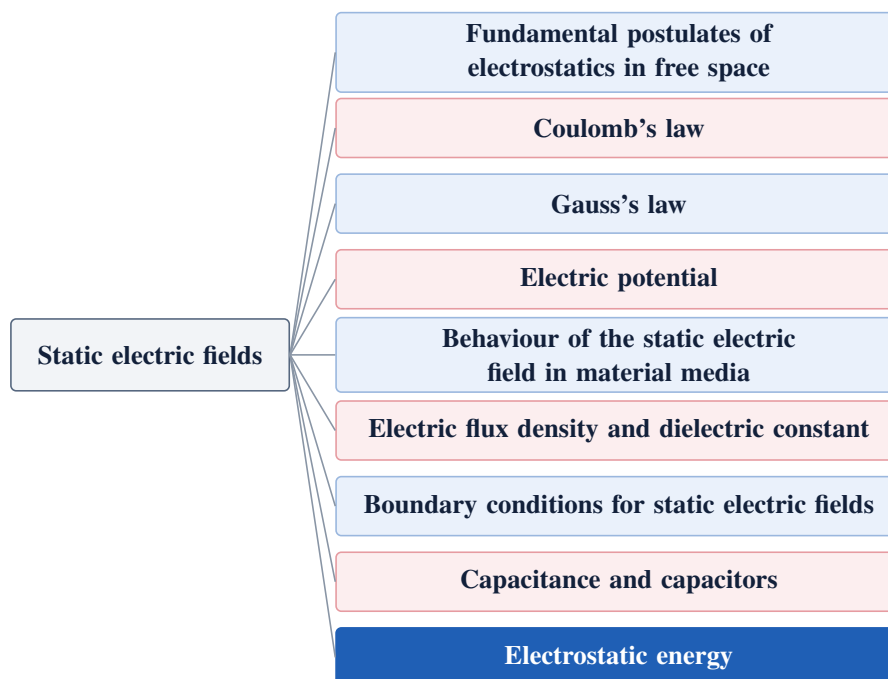
- Computing the potential difference between the conductors in terms of Q

$$V = - \int_{R_o}^b \frac{Q}{4\pi\epsilon_0(2\epsilon_r)R^2} dR - \int_b^{R_i} \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} dR = \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o} \right)$$

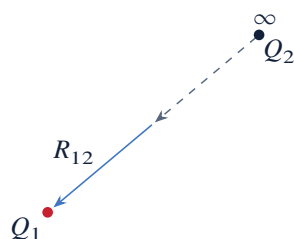
- Determining the capacitance

$$C = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o}}$$

2.15 Electrostatic Energy



- The electric potential of a point in an electric field is equal to the work needed to bring a unit positive charge from infinity (the zero-potential reference) to that point.
- To bring a charge Q_2 from infinity to the distance R_{12} from the charge Q_1 , the work needed is



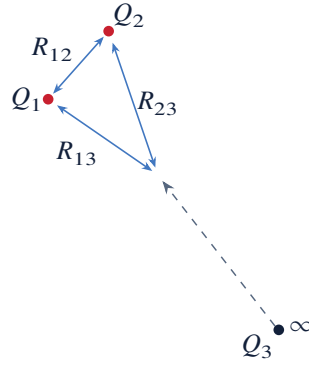
$$W_2 = Q_2 V_2 = Q_2 \frac{Q_1}{4\pi\epsilon_0 R_{12}}$$

- V_2 : the potential at the location of the charge Q_2

$$W_2 = Q_1 \frac{Q_2}{4\pi\epsilon_0 R_{12}} = Q_1 V_1$$

$$W_2 = \frac{1}{2}(Q_1 V_1 + Q_2 V_2)$$

- This work is stored in the system of two charges as potential energy.
- Now suppose another charge Q_3 is brought from infinity to a point at the distance R_{13} from Q_1 and R_{23} from Q_2 ; the additional work needed is



$$\Delta W = Q_3 V_3 = Q_3 \left(\frac{Q_1}{4\pi\epsilon_0 R_{13}} + \frac{Q_2}{4\pi\epsilon_0 R_{23}} \right)$$

- V_3 : the potential at the location of the charge Q_3
- Thus the potential energy stored in the system of three charges is

$$\begin{aligned} W_3 &= W_2 + \Delta W = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1 Q_2}{R_{12}} + \frac{Q_1 Q_3}{R_{13}} + \frac{Q_2 Q_3}{R_{23}} \right) \\ W_3 &= \frac{1}{2} \left[Q_1 \left(\frac{Q_2}{4\pi\epsilon_0 R_{12}} + \frac{Q_3}{4\pi\epsilon_0 R_{13}} \right) + Q_2 \left(\frac{Q_1}{4\pi\epsilon_0 R_{12}} + \frac{Q_3}{4\pi\epsilon_0 R_{23}} \right) \right. \\ &\quad \left. + Q_3 \left(\frac{Q_1}{4\pi\epsilon_0 R_{13}} + \frac{Q_2}{4\pi\epsilon_0 R_{23}} \right) \right] = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3) \end{aligned}$$

- Thus the potential energy of N discrete point charges at rest is

⚡ Important result

$$W_e = \frac{1}{2} \sum_{k=1}^N Q_k V_k \quad (\text{J})$$

- where V_k is the electric potential at the point of Q_k due to the other charges.
- The SI unit of energy is the joule, which is a large unit. Here another unit, the electron-volt (eV), is used.
 - One electron-volt is the energy or work needed to move an electron against a potential difference of one volt.

$$1 \text{ (eV)} = (1.60 \times 10^{-19}) \times 1 = 1.60 \times 10^{-19} \quad (\text{J})$$

- In a continuous charge distribution of density ρ the stored energy is

⚡ Important result

$$W_e = \frac{1}{2} \int_{V'} \rho V dv \quad (\text{J})$$

- ▶ where V is the potential at the point where the volume charge density is ρ , and V' is the volume of the region in which ρ exists.

Example 2-21

Find the energy required to assemble a uniformly charged sphere of radius b and volume charge density ρ .

▶ **Solution.**

$$\begin{aligned}
 W_e &= \frac{\rho}{2} \int_{V'} V dv = \frac{\rho}{2} \int_0^b V 4\pi R^2 dR \\
 \mathbf{E}_{R1} &= \mathbf{a}_R \frac{Q}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho b^3}{3\epsilon_0 R^2}, \quad R \geq b \\
 \mathbf{E}_{R2} &= \mathbf{a}_R \frac{Q_R}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho R}{3\epsilon_0}, \quad 0 < R \leq b \\
 V &= - \int_{\infty}^R \mathbf{E} \cdot d\mathbf{R} = - \left[\int_{\infty}^b E_{R1} dR + \int_b^R E_{R2} dR \right] \\
 &= - \left[\int_{\infty}^b \frac{\rho b^3}{3\epsilon_0 R^2} dR + \int_b^R \frac{\rho R}{3\epsilon_0} dR \right] \\
 &= \frac{\rho}{3\epsilon_0} \left(b^2 + \frac{b^2}{2} - \frac{R^2}{2} \right) = \frac{\rho}{3\epsilon_0} \left(\frac{3}{2}b^2 - \frac{R^2}{2} \right) \\
 W_e &= \frac{\rho}{2} \int_0^b \frac{\rho}{3\epsilon_0} \left(\frac{3}{2}b^2 - \frac{R^2}{2} \right) 4\pi R^2 dR = \frac{4\pi\rho^2 b^5}{15\epsilon_0}
 \end{aligned}$$

- ▶ Earlier we obtained the electrostatic energy of a charge distribution in terms of the charge density and the potential.
- ▶ Sometimes it is better to express this energy in terms of field quantities.
- ▶ It can be shown that

Important result

$$W_e = \frac{1}{2} \int_{V'} \mathbf{D} \cdot \mathbf{E} dv \quad (\text{J})$$

$$W_e = \frac{1}{2} \int_{V'} \epsilon E^2 dv \quad (\text{J})$$

$$W_e = \frac{1}{2} \int_{V'} \frac{D^2}{\epsilon} dv \quad (\text{J})$$

► V' : all space

Example 2-22

Solve Example 2-21 using the expression of the energy in terms of field quantities.

► **Solution.**

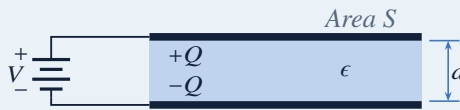
$$\mathbf{E}_{R1} = \mathbf{a}_R \frac{Q}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho b^3}{3\epsilon_0 R^2}, \quad R \geq b$$

$$\mathbf{E}_{R2} = \mathbf{a}_R \frac{Q_R}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho R}{3\epsilon_0}, \quad 0 < R \leq b$$

$$\begin{aligned} W_e &= \frac{1}{2} \epsilon_0 \int_{V'} |\mathbf{E}|^2 dv = \frac{1}{2} \epsilon_0 \int_0^b \left(\frac{\rho R}{3\epsilon_0} \right)^2 4\pi R^2 dR \\ &\quad + \frac{1}{2} \epsilon_0 \int_b^\infty \left(\frac{\rho b^3}{3\epsilon_0 R^2} \right)^2 4\pi R^2 dR = \frac{4\pi \rho^2 b^5}{15\epsilon_0} \end{aligned}$$

Example 2-23

In the figure below a parallel-plate capacitor of area S and plate separation d is charged by the voltage V . The permittivity of the insulator is ϵ . Find the stored energy.



► **Solution.**

$$E = \frac{V}{d}$$

$$W_e = \frac{1}{2} \int_{V'} \epsilon \left(\frac{V}{d} \right)^2 dv = \frac{1}{2} \epsilon \left(\frac{V}{d} \right)^2 (Sd) = \frac{1}{2} \left(\epsilon \frac{S}{d} \right) V^2$$

⚡ Important result

$$W_e = \frac{1}{2} CV^2 \quad (\text{J}) \qquad W_e = \frac{1}{2} QV \quad (\text{J}) \qquad W_e = \frac{Q^2}{2C} \quad (\text{J})$$

📌 Remark

It can be proved that the above equations hold for any capacitor made of two conductors.

Example 2-24

Using the energy relation, find the capacitance of a cylindrical capacitor of length L consisting of an inner conductor of radius a and an outer conductor of radius b , with permittivity ϵ .

► **Solution.**

$$\begin{aligned}\mathbf{E} &= \mathbf{a}_r E_r = \mathbf{a}_r \frac{Q}{2\pi\epsilon L r}, \quad a < r < b \\ W_e &= \frac{1}{2} \int_a^b \epsilon \left(\frac{Q}{2\pi\epsilon L r} \right)^2 (L 2\pi r dr) \\ &= \frac{Q^2}{4\pi\epsilon L} \int_a^b \frac{dr}{r} = \frac{Q^2}{4\pi\epsilon L} \ln \frac{b}{a} \\ \frac{Q^2}{2C} &= \frac{Q^2}{4\pi\epsilon L} \ln \frac{b}{a} \quad \Rightarrow \quad C = \frac{2\pi\epsilon L}{\ln \frac{b}{a}}\end{aligned}$$

